

An Independent Reproduction of the AYO ExoEarth Yield for the Habitable Worlds Observatory: Two Questions on Which It Turns and an application to a mass-selected planet sample in a narrowed habitable zone

Prepared in the `yields` vaibify project

September 26, 2026

Abstract

Yield estimates for the Habitable Worlds Observatory (HWO) rest on the Altruistic Yield Optimization (AYO) framework of Stark et al., and the exoEarth candidate (EEC) counts that flow from it now carry architectural weight. We report an independent reimplementa- tion of that model, built from the published equations and parameter tables rather than from AYO itself, and use it to evaluate a counterfactual planet selection: planets of $0.5\text{--}2\,M_{\oplus}$ with $R = M^{0.27}$ in a habitable zone of $0.96\text{--}1.20$ AU scaled as $\sqrt{L_{\star}}$, against the canonical $R \leq 1.4\,R_{\oplus}$ planets in $0.95\text{--}1.67$ AU. The reimplementa- tion follows every documented choice of the published method but one, and carries no parameter fitted to the published results: the 6 m yield is 20.92 against a published 22.5. Of 24 independent comparisons against the published record, 21 agree, and the realized-yield distribution lies within a total-variation distance of 0.037 of the published one (mean 22.4 against 21.0, median 19 against 18).

Two disagreements survive every attempt we have made to remove them, and we argue that both are under-determination in the published description rather than defects we can locate. First, spectral characterization times in our model depend on target difficulty and aperture far more steeply than Stark’s: the mean over the first eighteen candidates is 14.3 d at 6 m against a published 22 d. A 56-point factorial design over the four throughput-family parameters that could plausibly absorb this shows that *no* point in that family reaches the published value: among the 20 design points that reproduce the published yield to 10%, the characterization time spans only 7.0–11.6 d. Second, the $\eta_{\oplus}$ interval of Stark et al. (2024) Sec. 3.6 admits two readings, and a 2×2 decomposition attributes 99.1% of our residual disagreement with their Fig. 10 to that single choice. Following every documented choice simultaneously makes the agreement *worse*, not better (peak 15 against 10).

Each discrepancy reduces to one question that only the AYO authors can answer, and we state both explicitly. Until they are answered the reproduction is an “almost”, and we quantify what “almost” costs. The counterfactual result is meanwhile robust to both: the yield ratio is $0.232^{+0.027}_{-0.021}$ (68% interval), and across seven model variants spanning both readings it stays between 0.2201 and 0.2386. Stark’s own per-visit albedo test, which we adopt, gives an albedo penalty of 3.6% rather than the 12% he reports.

Contents

1	Introduction	3
2	An independent implementation of AYO	3
2.1	Scope and sources	3

2.2	The model as a graphical model	4
2.3	The nodes	4
2.4	What we do not calibrate	4
3	The model in equations	6
3.1	Stars	6
3.2	Occurrence and the selection boxes	7
3.3	Injected planets and their geometry	7
3.4	Count rates	8
3.5	The extended-source throughput T_{sky}	8
3.6	Exposure times	9
3.7	Counting, completeness and the albedo test	10
3.8	The survey allocation	11
3.9	Exozodi, $\eta_{\oplus}$ and the yield distribution	11
4	Agreement with the published record	11
5	First discrepancy: the characterization time budget	12
5.1	The symptom	12
5.2	The throughput family cannot supply the difference	12
5.3	What this implies	13
6	Second discrepancy: the reading of the occurrence interval	16
6.1	One sentence, two readings	16
6.2	A factorial decomposition	16
6.3	Following every documented choice makes the agreement worse	17
6.4	Testing the interval against its own source	17
7	The counterfactual, and its robustness	18
8	Discussion: the two questions	18
9	Conclusions	20
A	Verification against the published record	21
B	Published figures beside their model counterparts	21
C	Ledger of hypotheses tested	21
D	Every contributing star	21

1 Introduction

The Astro2020 decadal survey recommended a mission capable of surveying ~ 25 potentially habitable worlds. Translating that recommendation into an aperture, a coronagraph and a set of detector requirements runs through a yield model, and for HWO that model is AYO (Stark et al. 2014; Stark et al. 2019; Stark et al. 2024). AYO is not a simple scaling: it allocates exposure time across a target list by an equal-slope rule, so its output responds to changes in the planet population in ways a proportionality argument does not capture. When the expected number of detections falls, characterization time is freed and the survey reallocates; the yield is therefore a sublinear function of the occurrence rate, and a counterfactual about the planet population cannot be answered by rescaling a published number.

This has a practical consequence for anyone outside the AYO development team who wants to ask such a counterfactual: they must reimplement the model. That is what we did. Our target question was a mass-selected sample in a narrowed habitable zone (Sec. 7), but the reimplementation is the substantive part of the work, and its result is of wider interest than the counterfactual that motivated it, because it is a test of whether the published description of AYO is sufficient to reconstruct AYO.

Our finding is that it very nearly is. The model we built from the papers agrees with the published record on 21 of 24 independent comparisons, reproduces the shape of the published yield distribution, and lands within 7.0% of the headline 6 m yield with no parameter fitted to it. Where the published text describes a method, we follow it, even where a substitute would agree better with a published number. That level of agreement is not the interesting part of this paper. The interesting part is the residue: two disagreements that we have been unable to remove, that are not independent of each other in their consequences, and that between them control the quantities a mission designer would care most about — the target mix, the characterization time budget, and the width of the yield distribution.

We argue that neither residue is a bug in our implementation. Each traces to a point where the published description admits more than one reading, and in each case we can show that the readings are distinguishable, that the published numbers do not settle which is intended, and that the choice materially changes the answer. Sections 5 and 6 present the evidence for that claim. Section 8 states the two questions and argues that the reproduction hinges on them.

We have deliberately not closed either gap by tuning. In particular, the model carries no throughput calibration factor fitted to the published 6 m yield, for reasons set out in Sec. 2.4 that are themselves an argument about reproducibility: a free multiplier fitted to the quantity one is trying to reproduce does not merely inflate the agreement, it conceals every other error whose signature is a near-uniform change in exposure time.

2 An independent implementation of AYO

2.1 Scope and sources

We reimplemented the detection and characterization model of Stark et al. (2019) and the survey construction of Stark et al. (2014), with the mission parameters, coronagraph performance and astrophysical distributions of Stark et al. (2024). Nothing is taken from the AYO codebase, which is not public; where a quantity is published only as a figure we digitized the vector content stream of the e-print rather than reading values by eye, a decision forced on us early when six values read off a coronagraph figure proved wrong by up to a factor of three.

The target sample is the real HPIC (Tuchow et al. 2024). The coronagraph is the DMVC6 design, with core throughput and raw contrast digitized from the published curves. Exozodi is drawn from the LBTI HOSTS maximum-likelihood distribution digitized from Stark et al. (2024) Fig. 9, with four stars pinned to their measured values. Planetary albedos are drawn from the published uniform

distribution and counted with Stark et al. (2024) Sec. 3.2’s per-visit flux test, including its upper bound set by the inner working angle. The occurrence prior follows SAG13, normalized to the $\eta_{\oplus}$ of Stark et al. (2024) Sec. 3.6 — a choice that becomes the subject of Sec. 6.

2.2 The model as a graphical model

Writing the pipeline down as a probabilistic graphical model (Fig. 1, Table 1) is not decoration. It makes one distinction precise that the yield literature often leaves implicit: which uncertainties reach the survey’s decision node, and which do not. An uncertainty that reaches the allocation is *actionable* — knowing it before launch would change what the survey does — and one that does not is merely a spread on the outcome. $\eta_{\oplus}$ and the exozodi levels z_i reach the allocation; the per-planet albedo A_{ij} and the per-star planet count N_i do not. That distinction is what makes the counterfactual of Sec. 7 a re-optimization rather than a rescaling.

2.3 The nodes

Node	Type	Specification	Source; step
$\eta_{\oplus}$	random, global	lognormal, median 0.257, ln-width 0.76 (Sec. 6)	Stark+2024 Sec. 3.5, read at $z = 1$; A06
α, β	random, global	SAG13 shape; posterior of an <code>emcee</code> chain, used only through $\rho_{\mathcal{B}}$	SAG13; A06
$\rho_{\mathcal{B}}$	deterministic	$\eta_{\mathcal{B}}/\eta_{\oplus}$, the SAG13 integral over $\mathcal{B}$ relative to the canonical box; independent of Γ	A06
$\eta_{\mathcal{B}}$	deterministic	$\eta_{\oplus} \rho_{\mathcal{B}}$	A06
$\mathbf{s}_i$	observed	$d, L_*, T_{\text{eff}}, R_*$, HIP id	HPIC (Tuchow+2024); A01
z_i	random, per star	exozodi level, $z_i \sim p_z$; four stars fixed at LBTI values; 200 joint draws of $\mathbf{z}$	Stark+2024 Sec. 3.3, Fig. 9; A02, A04
C_i	deterministic	planner’s completeness $C_i(\tau, k)$ at $A_G = 0.2$, over exposure and visits, counting only planets that are also characterizable	Stark+2019; A04
λ	deterministic	equal-slope multiplier closing the two-year budget	Stark+2014; A05
$\mathbf{a}_i$	decision	(τ_i, k_i) : the vertex of star i ’s concave cost envelope at slope λ	A05, A07
N_i	random, per star	$N_i \sim \text{Poisson}(\eta_{\mathcal{B}})$	Stark+2024 Sec. 3.1; A07
$\mathbf{o}_{ij}$	random, per planet	circular orbit: a, R from SAG13 over $\mathcal{B}$; $\cos i \sim \mathcal{U}$; independent epoch per visit	ExoVista/AYO; A04
A_{ij}	random, per planet	geometric albedo, $\mathcal{U}(0.08, 0.32)$, counted by Stark’s per-visit flux test	Stark+2024 Sec. 3.2; A02, A04
τ_{ij}	deterministic	detection time per visit (two channels in quadrature, S/N 7) and characterization time (R 140, best of nine bandpasses, best orbital phase)	Stark+2019 Eqs. 1–9; A04
D_{ij}	deterministic	$\mathbb{K}[\min_{v \leq k_i} \tau_{ijv}^{\text{det}} \leq \tau_i \wedge \tau_{ij}^{\text{char}} \leq 2 \text{ mo}]$	Stark+2019 Sec. 3; A04
Y	deterministic	$\sum_{i,j} D_{ij}$	A07

Table 1: Every node of Fig. 1, what it is, and where it comes from.

2.4 What we do not calibrate

A common way to absorb the unpublished parts of a model is a scalar throughput factor κ on every band, fitted so that the canonical box reproduces the published 22.5 candidates. We do not use one, for two reasons. First, no Stark paper contains such a factor: 22.5 is an output of AYO, not an input to it, so a reproduction that fits to it is not reproducing the same quantity. Second, a fitted factor hides errors. We measured the effect of every alternative treatment we tested on the per-planet exposure times, and found that with κ refitted each of them changes those times by a nearly uniform factor of 1.0 ± 0.05 . Because the yield responds to throughput with an elasticity of only ~ 0.4 , a fitted factor can absorb any such change completely, and a whole class of modeling

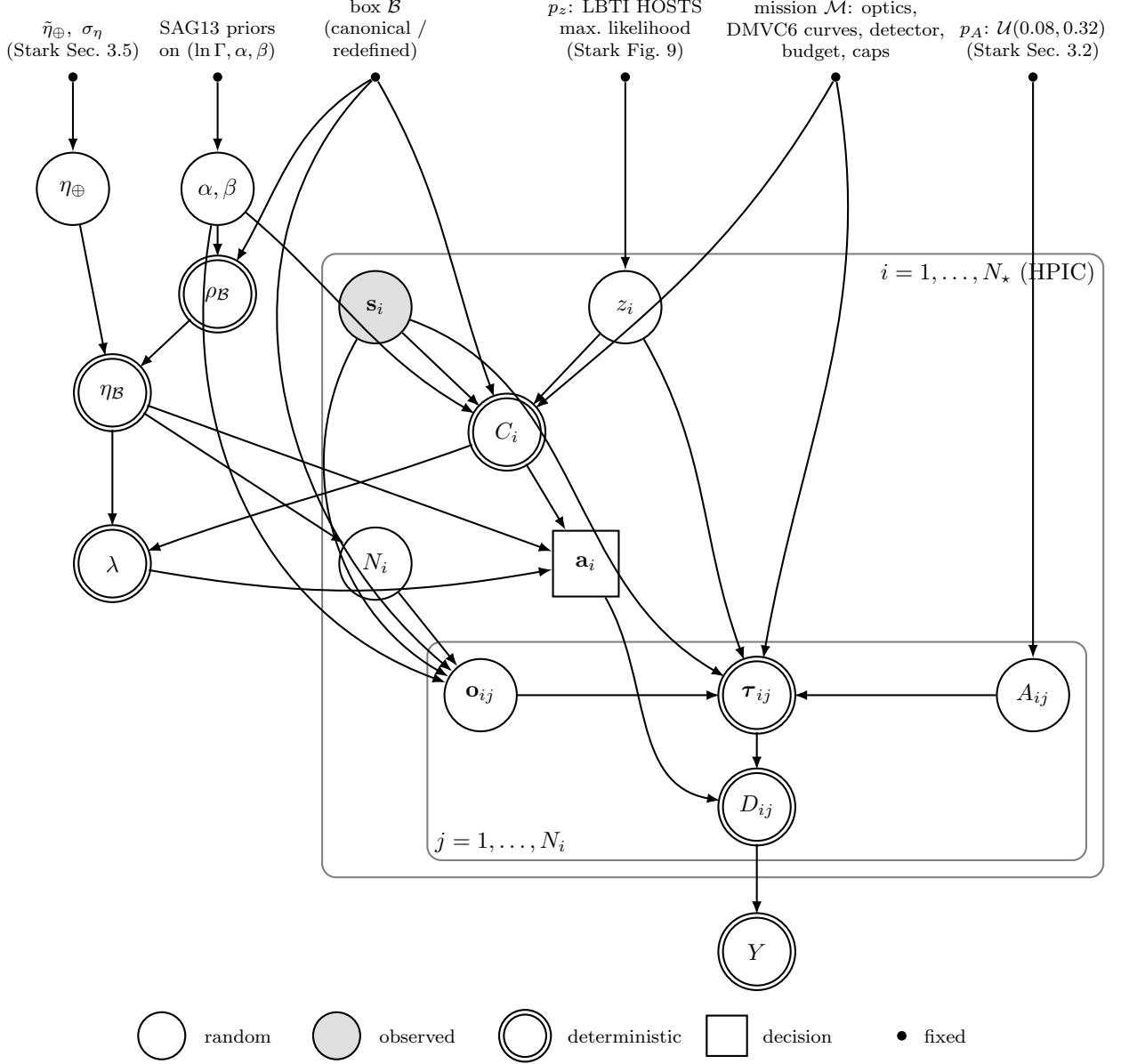


Figure 1: The yield model as an influence diagram. Outer plate: the $N_{\star}$ screened HPIC stars; inner plate: the N_i planets actually present around star i . The survey allocation $\mathbf{a}_i = (\tau_i, k_i)$ — exposure per visit and number of visits — is the one decision node; it is set by the equal-slope multiplier λ from the planner’s completeness curves C_i and the box occurrence rate $\eta_{\mathcal{B}}$. A planet is counted ($D_{ij} = 1$) when its detection time at some visit is within the allocation and its spectral characterization time is within the two-month cap; $Y = \sum_{ij} D_{ij}$. Four stars with LBTI-measured exozodi have z_i observed rather than drawn (not shown). The edges that decide the uncertainty budget are the ones *into* $\mathbf{a}_i$: $\eta_{\oplus}$ and z_i reach it (actionable), A_{ij} and N_i do not (static).

errors becomes invisible. Without it, the two residues of Secs. 5 and 6 are visible as residues rather than silently rescaled away. For reference, the factor that would reproduce 22.5 is 1.20.

One sourced term matters a great deal here: the extended-source throughput T_{sky} that Stark et al. (2019) Eqs. 5–6 apply to the zodiacal and exozodiacal backgrounds. Its radial shape is not published, so the reconstruction is ours, and it very probably differs from AYO’s near the inner working angle. Section 3.5 gives our reconstruction, what AYO’s is likely to be, and by how much they differ.

3 The model in equations

This section states every equation the pipeline evaluates, in the order it evaluates them, with each variable defined where it first appears. Table 2 lists the scalar parameters with their values and sources, and Table 3 the detection channels and spectral bandpasses. Throughout, D is the inscribed diameter by which Stark et al. (2024) label the design, and $D_{\text{circ}} = 7.16$ m the circumscribed diameter against which the published coronagraph curves are plotted.

Parameter	Value	Source
Inscribed diameter D (m)	6.0	S24
Circumscribed / inscribed diameter	1.194	LUVOIR-B, 8 m / 6.7 m
Area fill factor f_{fill}	0.785	none published
Contamination throughput τ_{con}	0.95	S24
Quantum efficiency Q	0.90	S24
Detective quantum efficiency q	0.75	S24
Dark current ξ_{DC} (counts pix ⁻¹ s ⁻¹)	3.0×10^{-5}	S24
Read noise (counts pix ⁻¹)	0	S24 (photon counting)
Clock-induced charge ξ_{CIC} (counts pix ⁻¹ frame ⁻¹)	1.3×10^{-3}	S24
Raw-contrast floor ζ_{floor}	10^{-10}	S19
Photometric aperture radius X (λ/D)	0.7	S19
Noise floor Δm_{nf} (mag)	26.5	S24
Planning albedo A_{plan}	0.20	S24 Sec. 3.2
Exozodi surface brightness per zodi at V (mag arcsec ⁻²)	22.0	S14
Survey time T (yr)	2.0	S24
Exposure limit t_{lim} , including overheads (d)	60	S24
Slew overhead t_{slew} (h)	1.0	S24
Wavefront-control overhead t_{WFC} (h)	2.7	S24
Wavefront-control time multiplier τ'	1.10	S24
Maximum visits per star k_{max}	6	S15
Circumscribed diameter D_{circ} (m)	7.16	derived
Collecting area $A = f_{\text{fill}} \pi D_{\text{circ}}^2 / 4$ (m ²)	31.6	derived
Longest science exposure τ_{cap} (d), Eq. (18)	54.4	derived
Airy encircled energy in the aperture, $\text{EE}(X)$	0.678	derived
Peak DMVC6 core throughput $\Upsilon_{\text{c,max}}$	0.388	digitized

Table 2: Scalar parameters of the baseline mission. S14, S15, S19 and S24 are Stark et al. (2014, 2015, 2019, 2024); S24 values are from its Tables 1 and 2 unless a section is given. The DMVC6 core throughput $\Upsilon_{\text{c}}(r)$ and raw contrast $\zeta(r)$ are digitized from Stark et al. (2024) and shown in Fig. 15.

3.1 Stars

Star i has distance d_i , luminosity L_i , effective temperature T_i , radius $R_{\star,i}$ and ecliptic latitude b_i , all from HPIC. Its photon spectral flux at the telescope is a blackbody,

$$F_{\star}(\lambda) = \pi \left(\frac{R_{\star}}{d} \right)^2 B_{\lambda}(T) \frac{\lambda}{hc} \quad [\text{photons s}^{-1} \text{ m}^{-2} \text{ m}^{-1}], \quad (1)$$

where B_{λ} is the Planck function. Its Earth-equivalent insolation distance is $a_{\text{EEID}} = \sqrt{L/L_{\odot}}$ AU.

Name	Use	λ (nm)	$\Delta\lambda/\lambda$	τ_b	S/N	N_{pix}
LW	detection	550	0.20	0.34	7	4
SW	detection	450	0.20	0.15	7	4
IFS730	spectrum	730	1/140	0.23	17	51
IFS763	spectrum	763	1/140	0.23	15	56
IFS779	spectrum	779	1/140	0.23	14	58
IFS824	spectrum	824	1/140	0.23	13	65
IFS842	spectrum	842	1/140	0.23	12	68
IFS917	spectrum	917	1/140	0.23	10	81
IFS930	spectrum	930	1/140	0.23	9	83
IFS943	spectrum	943	1/140	0.23	6	85
IFS1000	spectrum	1000	1/140	0.23	5	96

Table 3: Detection channels and characterization bandpasses. τ_b is the optical throughput of the channel, S/N the required signal-to-noise ratio (per spectral bin for the spectra), and N_{pix} the number of detector pixels the signal is spread over. The two detection channels observe in parallel and their S/N adds in quadrature; a spectrum uses whichever single bandpass is fastest. The spectral bandpasses and their S/N follow Stark et al. (2024) Fig. 3.

3.2 Occurrence and the selection boxes

Planets follow the SAG13 density

$$\frac{d^2 N}{d \ln R d \ln P} = \Gamma R^\alpha P^\beta, \quad (2)$$

with R in Earth radii and P in years. A selection box $\mathcal{B}$ is a range of scaled semi-major axis $a' = a/a_{\text{EEID}} \in [a'_{\text{in}}, a'_{\text{out}}]$ and a radius range $[R_{\text{lo}}(a'), R_{\text{hi}}]$. Since $P \propto a^{3/2}$ at the solar-twin reference, the occurrence over the box is

$$\eta_{\mathcal{B}} = \frac{3}{2} \Gamma \int_{a'_{\text{in}}}^{a'_{\text{out}}} a'^{3\beta/2} \int_{R_{\text{lo}}(a')}^{R_{\text{hi}}} R^\alpha d \ln R d \ln a'. \quad (3)$$

The canonical box has $a' \in [0.95, 1.67]$, $R_{\text{lo}} = 0.8 a'^{-1/2}$ and $R_{\text{hi}} = 1.4$. The redefined box has $a' \in [0.96, 1.20]$ and $0.5 \leq M/M_\oplus \leq 2$ with $R = M^{0.27}$, i.e. $R \in [0.829, 1.206]$. The ratio $\rho_{\mathcal{B}} = \eta_{\mathcal{B}}/\eta_{\text{canon}}$ does not depend on Γ ; it is computed over an `emcee` chain in $(\ln \Gamma, \alpha, \beta)$ with SAG13 priors $\alpha = -0.19 \pm 0.27$ and $\beta = 0.26 \pm 0.29$.

The canonical occurrence $\eta_\oplus \equiv \eta_{\text{canon}}$ is lognormal,

$$\ln \eta_\oplus \sim \mathcal{N}(\mu, \sigma^2), \quad \mu = \frac{1}{2} \ln(\eta_{\text{lo}} \eta_{\text{hi}}), \quad \sigma = \frac{\ln(\eta_{\text{hi}}/\eta_{\text{lo}})}{2z}, \quad (4)$$

where $[\eta_{\text{lo}}, \eta_{\text{hi}}] = [0.12, 0.55]$ is the interval quoted by Stark et al. (2024) and z is the number of standard deviations it is taken to span. We adopt $z = 1$ (a 68% interval), which gives a median $e^\mu = 0.257$, $\sigma = 0.76$ and a mean of 0.343; Sec. 6 explains and tests this choice. The redefined box then has $\eta_{\mathcal{B}} = \rho_{\mathcal{B}} \eta_\oplus$.

3.3 Injected planets and their geometry

For each star and box, $N_{\text{inj}} = 1\,200\text{--}3\,000$ planets are drawn: a' from a density $\propto a'^{3\beta/2} d \ln a'$ on the box's range, R from $\propto R^\alpha d \ln R$ on its radius range (with α and β at their SAG13 central values), $\cos i$ uniform on $[-1, 1]$, and, for each of up to k_{max} visits v , an independent orbital angle θ_v uniform on $[0, 2\pi)$. Orbits are circular with radius $a = a' a_{\text{EEID}}$. At visit v the projected separation and phase angle ϕ are

$$s_v = a \sqrt{\cos^2 \theta_v + \sin^2 \theta_v \cos^2 i}, \quad \cos \phi_v = \sin \theta_v \sin i, \quad (5)$$

and the planet-to-star flux ratio is

$$f_v = A \Phi(\phi_v) \left(\frac{R_p}{a} \right)^2, \quad \Phi(\phi) = \frac{\sin \phi + (\pi - \phi) \cos \phi}{\pi}, \quad (6)$$

with A the geometric albedo: $A_{\text{plan}} = 0.2$ for planning and a draw from $\mathcal{U}(0.08, 0.32)$ for counting (Sec. 3.7). The separation in the units of the coronagraph curves is

$$r_v = \frac{s_v/d}{\lambda/D} \cdot \frac{D_{\text{circ}}}{D}, \quad (7)$$

with s_v/d in radians.

3.4 Count rates

For a channel b with central wavelength λ , bandwidth $\Delta\lambda$, optical throughput τ_b and N_{pix} pixels, the end-to-end throughput is $\tau = \tau_b \tau_{\text{con}} Q q$ and the stellar count rate is $\text{CR}_\star = F_\star(\lambda) \Delta\lambda A_{\text{tel}} \tau$, where $A_{\text{tel}} = 31.6 \text{ m}^2$ is the collecting area (Table 2). The coronagraph enters through the core throughput $\Upsilon_c(r)$ and the raw contrast $\zeta(r) = \max[\zeta_{\text{DMVC6}}(r), \zeta_{\text{floor}}]$. With $\Omega = \pi(X\lambda/D)^2$ the solid angle of the photometric aperture of radius X , the count rates are

$$\text{CR}_p = \text{CR}_\star \Upsilon_c(r) f, \quad (8)$$

$$\text{CR}_{\text{leak}} = \text{CR}_\star \Upsilon_c(r) \zeta(r), \quad (9)$$

$$\text{CR}_z = I_z \Delta\lambda \Omega A_{\text{tel}} \tau T_{\text{sky}}(r), \quad (10)$$

$$\text{CR}_{ez} = z I_{ez} \Delta\lambda \Omega A_{\text{tel}} \tau T_{\text{sky}}(r), \quad (11)$$

$$\text{CR}_{\text{det}} = N_{\text{pix}} \left[\xi_{\text{DC}} + 6.73 \xi_{\text{CIC}} \frac{\text{CR}_p + \text{CR}_{\text{leak}} + \text{CR}_z + \text{CR}_{ez}}{N_{\text{pix}}} \right]. \quad (12)$$

Here I_z is the local zodiacal surface brightness, $I_z = 0.69 I_{90}(\lambda) f_{135}(b)$, with I_{90} the Leinert et al. (1998) spectrum at 90° elongation and $f_{135}(b) = 1.02 - 0.566x - 0.884x^2 + 0.853x^3$, $x = |\sin b|$ (Stark et al. 2014); z is the star’s exozodi level in zodis; and

$$I_{ez} = F_0(\lambda) 10^{-0.4 \times 22} \frac{F_\star(\lambda)|_{10 \text{ pc}}}{F_\odot(\lambda)|_{10 \text{ pc}}} \frac{L_\odot}{L} \quad (13)$$

is the surface brightness of one zodi of exozodiacal dust around that star, held at constant habitable-zone optical depth (Stark et al. 2014, App. C). $F_0(\lambda)$ is a solar-colored zero point normalized to $10^{11} \text{ photons s}^{-1} \text{ m}^{-2} \mu\text{m}^{-1}$ at V . The detector term is Stark et al. (2019) Eq. 9: dark current plus clock-induced charge, whose frame rate is set by the brightest pixel, with the Geiger factor 6.73. The total background is $\text{CR}_b = \text{CR}_{\text{leak}} + \text{CR}_z + \text{CR}_{ez} + \text{CR}_{\text{det}}$.

3.5 The extended-source throughput T_{sky}

Only the two sky terms, Eqs. (10) and (11), carry T_{sky} . Stark et al. (2019, Eqs. 5–6) define it as “the instrument’s throughput for extended sources”, computed “by first convolving the spatially-dependent PSF at all locations with a normalized uniform background”. For a uniform background, the light an aperture at r collects is the sum, over every point of the sky nearby, of the fraction of that point’s light the coronagraph delivers into the aperture. $T_{\text{sky}}(r)$ therefore counts *all* transmitted light landing in the aperture, from the cores and the wings of the PSFs of every nearby point, while $\Upsilon_c(r)$ counts only the core of one point source. Stark’s DMVC6 map is not published, so it has to be reconstructed.

Our reconstruction. Write $\Upsilon_c(r) = T_{\text{tot}}(r) \epsilon_{\text{core}}(r)$, the total fraction T_{tot} of an off-axis source’s light that the coronagraph transmits times the fraction ϵ_{core} of that light falling inside the photometric aperture. If the off-axis PSF keeps the Airy shape at every separation, $\epsilon_{\text{core}} = \text{EE}(X)$, and if

T_{tot} changes little across one λ/D , the convolution returns T_{tot} itself. Under those two assumptions

$$T_{\text{sky}}(r) = \frac{\Upsilon_c(r)}{\text{EE}(X)} = 1.474 \Upsilon_c(r), \quad (14)$$

since $\text{EE}(0.7) = 0.678$. For the DMVC6, T_{sky} therefore rises from 0.08 at $2\lambda/D$ to 0.572 at the outer edge of the dark hole, in exact proportion to the core throughput (Table 4, Fig. 2).

What AYO probably does. AYO computes the convolution from its simulated two-dimensional off-axis PSFs, and both of our assumptions fail near the inner working angle. There the focal-plane mask distorts the off-axis PSF and moves light from its core into its wings, so ϵ_{core} falls below $\text{EE}(X)$; and the convolution averages T_{tot} over about one λ/D , pulling in light from larger separations where more is transmitted. Both raise $T_{\text{sky}}/\Upsilon_c$ toward the inner working angle, where Eq. (14) holds it constant. The only AYO values in print are for a different coronagraph, the vortex of the Stark et al. (2025) exposure-time benchmark. There AYO’s $T_{\text{sky}}/\Upsilon_c$ rises from 1.8 at $5.7\lambda/D$ to 3.2 at $2.1\lambda/D$, and Eq. (14) applied to that coronagraph gives only 0.47–0.81 of AYO’s value. We condense the benchmark into a correction factor linear in the core fraction $u = \Upsilon_c/\Upsilon_{c,\text{max}}$,

$$T_{\text{sky}}^{\text{AYO-like}}(r) = T_{\text{sky}}(r) [F_0 + (F_1 - F_0)u], \quad F_0 = 2.19, \quad F_1 = 1.08, \quad (15)$$

which fits the benchmark to 6% rms. Applied to the DMVC6, this AYO-like shape exceeds ours by a factor of 2.04 at $2\lambda/D$, 1.52 at $5\lambda/D$ and 1.26 at $10\lambda/D$ (Table 4).

r (λ/D_{circ})	Υ_c	T_{sky} , Eq. (14)	T_{sky} , AYO-like	ratio
1.5	0.016	0.023	0.050	2.15
2.0	0.053	0.079	0.160	2.04
2.5	0.097	0.143	0.274	1.91
3.0	0.134	0.197	0.357	1.81
3.5	0.168	0.247	0.423	1.71
4.0	0.195	0.288	0.470	1.63
5.0	0.235	0.347	0.527	1.52
7.0	0.286	0.421	0.579	1.38
10.0	0.325	0.480	0.605	1.26
15.0	0.357	0.527	0.616	1.17
20.0	0.373	0.550	0.619	1.12
25.0	0.382	0.564	0.619	1.10

Table 4: The DMVC6 core throughput and the two reconstructions of its extended-source throughput. The last column is the AYO-like value divided by ours.

Why this could matter, and what it does. T_{sky} multiplies the zodiacal and exozodiacal backgrounds, and at $R = 140$ near $1\mu\text{m}$ the exozodiacal term dominates the noise of a spectrum. The planets that sit closest to the inner working angle are those around distant and luminous stars, which are exactly the targets on which our survey and Stark’s disagree (Sec. 5). A larger T_{sky} near the inner working angle is therefore a natural candidate for the characterization-time discrepancy, and the term demonstrably matters for the absolute yield: holding T_{sky} at its large-separation value lowers the mean realized yield from 22.4 to 19.2 (Table 5). We test the AYO-like shape directly in Sec. 5, as one of the four parameters of the throughput study, and find that it moves the model along the same curve as every other throughput change rather than toward Stark’s published values (Fig. 3). A DMVC6 map that departs from ours by more than the vortex benchmark suggests cannot be excluded, and it is part of Question 1 (Sec. 8).

3.6 Exposure times

For each channel c define

$$a_c = \text{CR}_{p,c}^2, \quad v_c = \text{CR}_{p,c} + 2\text{CR}_{b,c}, \quad n_c = \zeta_{\text{nf}} \text{CR}_{\star,c} \Upsilon_c(r), \quad \zeta_{\text{nf}} = \frac{10^{-0.4\Delta m_{\text{nf}}} \sqrt{N_{\text{ch}}}}{(S/N)_d},$$

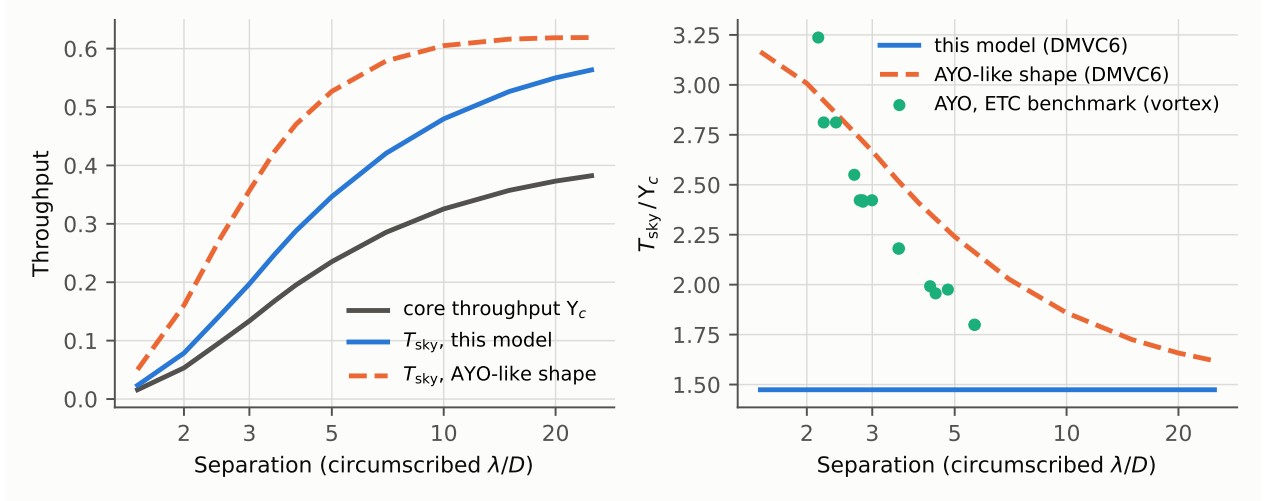


Figure 2: Left: DMVC6 core throughput and the two reconstructions of T_{sky} . Right: their ratio to the core throughput. Ours (Eq. (14)) is constant at 1.474; the AYO-like shape (Eq. (15)) rises toward the inner working angle, following AYO’s own values for the vortex coronagraph of the Stark et al. (2025) benchmark (points).

where n_c is the noise-floor rate and ζ_{nf} the contrast that reaches the detection S/N only in infinite time with all $N_{\text{ch}} = 2$ channels combined. The exposure time t that reaches signal-to-noise S/N solves

$$\sum_c \frac{a_c t}{v_c + n_c^2 t} = (S/N)^2, \quad (16)$$

which for a single channel is $t = (S/N)^2 v / [a - (S/N)^2 n^2]$, Stark et al. (2025) Eq. 1, and without the floor ($n = 0$) reduces to Stark et al. (2019) Eq. 1. A planet with $\sum_c a_c / n_c^2 \leq (S/N)^2$ can never be detected, and $t = \infty$. Detection uses the two channels of Table 3 at $S/N = 7$, giving t_{jv}^{det} for planet j at visit v . A spectrum uses one bandpass o at its own S/N; its time is minimized over the bandpasses and over 40 orbital angles θ , because the orbit is known by the time a spectrum is taken (Stark et al. 2019):

$$t_j^{\text{char}} = \min_{o, \theta} t_o(\theta). \quad (17)$$

The two-month limit applies to the exposure including overheads, so the longest science exposure is

$$\tau_{\text{cap}} = \frac{t_{\text{lim}} - t_{\text{slew}} - t_{\text{WFC}}}{\tau'} = 54.4 \text{ d}. \quad (18)$$

3.7 Counting, completeness and the albedo test

At planning albedo, planet j counts for a star observed k times with exposure t per visit if it is detected on at least one of those visits and its spectrum fits under the cap:

$$D_j(t, k) = \mathbb{1} \left[\min_{v \leq k} t_{jv}^{\text{det}} \leq \min(t, \tau_{\text{cap}}) \wedge t_j^{\text{char}} \leq \tau_{\text{cap}} \right], \quad (19)$$

and the star’s completeness is the counted fraction, $C(t, k) = N_{\text{inj}}^{-1} \sum_j D_j(t, k)$. The mean characterization time of the counted planets, $\bar{t}_{\text{char}}(t, k)$, is what the survey budgets for spectra.

To count planets with drawn albedos we follow Stark et al. (2024) Sec. 3.2: the plan stays fixed at A_{plan} , and a planet with drawn albedo A_j is counted at visit v only if the fixed plan would have caught it. Let $\mathcal{S}_v(t)$ be the set of planning-albedo planets detected at visit v within exposure t . The drawn planet passes at the smallest t for which its flux $f_{jv} A_j / A_{\text{plan}}$ is at least the faintest flux in $\mathcal{S}_v(t)$ and its separation is at least the smallest separation in $\mathcal{S}_v(t)$, and it fails outright if its flux exceeds the brightest flux any detectable planet reaches, which the inner working angle bounds.

Replacing t_{jv}^{det} in Eq. (19) by that threshold gives the completeness actually realized, $\bar{C}(t, k)$. Spectra are budgeted at A_{plan} , as Stark states.

3.8 The survey allocation

Visiting star i k times with exposure t costs

$$c_i(t, k) = k(\tau' t + t_{\text{oh}}) + \eta_{\mathcal{B}} C_i(t, k) [\tau' \bar{t}_{\text{char}, i}(t, k) + t_{\text{oh}}], \quad t_{\text{oh}} = t_{\text{slew}} + t_{\text{WFC}}, \quad (20)$$

detection for every visit plus a spectrum for every expected detection. Pooling all (t, k) options for star i gives a set of points (c_i, C_i) whose upper concave envelope $\mathcal{E}_i$ is the star’s best completeness for any time spent. For a slope λ , each star is allocated the last vertex of $\mathcal{E}_i$ whose incremental return $\Delta C_i / \Delta c_i$ is at least λ , and λ is set so that $\sum_i c_i = T$. This equal-slope rule maximizes the expected yield $Y_{\text{plan}} = \eta_{\mathcal{B}} \sum_i C_i$ under the budget (Stark et al. 2014) and matches an exact dynamic-programming optimum to 0.00%.

3.9 Exozodi, $\eta_{\oplus}$ and the yield distribution

Exozodi levels z_i are drawn from the HOSTS distribution, with four stars pinned to their LBTI values. Each star’s completeness is tabulated at 23 exozodi levels and interpolated to each of 200 joint draws of $\mathbf{z}$. For each posterior sample n of $\eta_{\oplus}$, paired with one exozodi draw, the survey is re-optimized and the expected yield is $\Lambda_n = \eta_{\mathcal{B}, n} \sum_i \bar{C}_i(t_i, k_i)$ at the allocated (t_i, k_i) . By Poisson thinning the realized yield is Poisson with that mean, so its distribution is the exact mixture

$$p(Y = y) = \frac{1}{N} \sum_{n=1}^N \frac{\Lambda_n^y e^{-\Lambda_n}}{y!}. \quad (21)$$

We compare it with Stark’s digitized Fig. 10 through the total-variation distance

$$D_{\text{TV}} = \frac{1}{2} \sum_{y=0}^{49} |p(y) - q(y)|, \quad (22)$$

where q is the published distribution and both are renormalized over the figure’s range $0 \leq y \leq 49$. D_{TV} is the largest difference in probability the two assign to any set of outcomes: 0 for identical distributions, 1 for disjoint ones.

4 Agreement with the published record

We hold ourselves to comparisons against numbers stated in the published *text*, and we tag as “tuned” any comparison against a quantity that entered our model as an input, so that it can never be counted as evidence. On that basis 21 of 24 comparisons agree (Appendix A). Nine of Stark’s figures are reproduced in their own axes alongside the originals (Appendix B).

Three agreements are worth singling out, because they are the ones that constrain the model most and because they frame the disagreements that follow.

The yield level. The uncalibrated 6 m expected yield is 20.92 against a published 22.5, a difference of 7.0% with nothing fitted.

The realized-yield distribution. Marginalizing over the occurrence posterior and Poisson counting reproduces the shape of the published Fig. 10 to a total-variation distance $D_{\text{TV}} = 0.037$ (Eq. 22): mean 22.4 against 21.0, median 19 against 18, $P(Y < 12) = 0.26$ against 0.29, and peak 12 against 10. We stress that the peak is a fragile statistic and should be computed as the exact Poisson mixture over the sampled expected yields rather than as the mode of a histogram of realized draws; the

latter estimator, bootstrapped, cannot distinguish 10 from 16, and a 5–10% change in the overall yield level moves the exact peak by one bin.

An independent implementation. EXOSIMS (Savransky & Garrett 2016), which shares no code and no scheduling method with either AYO or this work, agrees with our radiometry on identical stars and optics: local zodi 1.001, exozodi 0.975, stellar photon rate 0.979. EXOSIMS also *found* two of our errors, by carrying terms we lacked, each then confirmed against Stark’s own papers rather than against EXOSIMS.

The albedo penalty. Following Stark et al. (2024) Sec. 3.2, the observation plan is fixed at $A_G = 0.2$, and a planet with a drawn albedo counts only if its flux falls inside the range the visit actually reached: above the faintest flux the exposure detected, and below the brightest the inner working angle allowed. With that test the albedo penalty is 3.6%, against the 12% Stark reports; re-deriving each drawn-albedo planet’s exposure time instead gives 8.6%. Neither reproduces the published value, and we report the shortfall of Stark’s own method rather than substitute one that comes closer.

Against that background, the two sections that follow concern what did not come out right.

5 First discrepancy: the characterization time budget

5.1 The symptom

Our spectral characterization times are too short for easy targets and too long for hard ones. Averaged over the eighteen highest-priority candidates, we require 14.3 d at 6 m and 2.05 d at 9 m, against published values of 22 and 3.5 d. The ratio between apertures is therefore 7.0 against a published 6.2: our model’s characterization cost falls off far more steeply with aperture than Stark’s.

The same steepness appears in the target mix. Against Stark et al. (2024) Fig. 11 our survey allocates almost nothing to roughly half of the targets beyond 15 pc and above $2 L_\odot$. The mechanism is not mysterious in itself: at $R = 140$ near $1 \mu\text{m}$ the background is exozodi-dominated, the exozodi count rate is close to distance-independent, and the required time therefore scales as d^4 . An Earth twin needs 0.2 d at τ Cet and 255 d at 110 Her, against a two-month cap. What is mysterious is how AYO’s own survey reaches the completeness its figure shows for those stars, because AYO’s own exposure-time calculator gives comparable times: the Stark et al. (2025) ETC benchmark yields Earth-twin H_2O spectra of 37 d at 12.3 pc and 104 d at 18.2 pc, for a slightly better telescope than the baseline.

We have verified that our implementation of the exposure-time equations reproduces AYO’s own per-observation times to 0.94–1.02 when given AYO’s own inputs. The equations are not where we differ.

5.2 The throughput family cannot supply the difference

Before concluding that the discrepancy is irreducible we tested whether it could be absorbed by the parameters that plausibly could absorb it: a throughput calibration on all bands, a separate calibration on the spectral bands only, the normalization of the coronagraph leak term, and the radial shape of T_{sky} . These are the four levers that a reader might reasonably suspect we had set wrongly, and they are exactly the family that κ used to hide.

The four parameters, and the ranges we allow them, are:

- $\kappa \in [0.6, 3.0]$, a multiplier on the throughput of every band;
- $\kappa_c \in [0.4, 2.5]$, an extra multiplier on the spectral bands only, which can slow spectra down without touching detection;

- $f_{\text{leak}} \in [1, 1.78]$, the normalization of the leaked-starlight term, from our $\zeta \Upsilon_c$ to the $\zeta \text{PSF}_{\text{peak}} \Omega$ of Stark et al. (2019) Eq. 4 evaluated for an Airy pattern;
- $s_{\text{sky}} \in [0, 1]$, which interpolates T_{sky} from our reconstruction ($s_{\text{sky}} = 0$, Eq. 14) to the AYO-like shape ($s_{\text{sky}} = 1$, Eq. 15).

We evaluated a 56-point Latin hypercube over that box, running the full survey at each point at 6 and 9 m with exozodi fixed at three zodis, and recording the 6 m planning yield, the mean characterization time of the first eighteen candidates at 6 and 9 m, and the median target completeness in each distance bin of Fig. 11. Figure 3 shows every point against the published values, and Fig. 4 the posteriors of an emulator fitted to them. Four results follow.

- **No setting reaches the published characterization time.** This needs no model of the design: all 56 settings were run end to end, and the longest first-eighteen characterization time any of them produces at 6 m is 16.9 d, against the published 22 d. Even the settings that slow spectra the most, and in doing so cut the yield to about 10, stay below it (Fig. 3, top left).
- **At the published yield, spectra are about twice too fast.** A characterization time only means something alongside the yield it comes with, so we keep the 20 points whose 6 m yield is within 10% of 22.5 (the shaded band in Fig. 3). Their first-eighteen time spans 7.0–11.6 d at 6 m and 0.79–1.49 d at 9 m, against 22 and 3.5 d. Neither range comes near the published value, so matching the yield and matching the spectra are incompatible everywhere in the box.
- **Fitting the yield alone predicts the shortfall.** If the yield is the only constraint, as it is for a calibration factor, the emulator’s posterior has $\kappa = 1.39$ and $\kappa_c = 1.12$, strongly anti-correlated (correlation -0.80 between their logarithms), and it predicts a 6 m characterization time of 9.7 d (6.4–12.7). A pipeline calibrated to 22.5 would therefore have produced spectra about twice as fast as published without any sign of the problem.
- **The published observables pin down only one combination of parameters.** The singular values of the observables’ local sensitivity to the four parameters are 31.72, 2.31, 1.50, 0.90; the first is 13.7 times the second. Physically, all four act mostly as a single efficiency knob: anything that makes the instrument more efficient raises the yield and shortens every characterization time together. That is why the points in each panel of Fig. 3 fall on one narrow curve, whatever their T_{sky} shape. Reaching the published values would require moving off that curve, and no parameter in the box does so.

Characterization time *can* be bought inside this family, but only by giving up yield: when every published observable is imposed at once, the posterior pushes κ_c onto its lower bound trying to slow spectra down, and the yield falls to 16.4. Buying agreement that way is not a reproduction.

5.3 What this implies

We can state the disagreement sharply, which is what makes it a question rather than a complaint. Our exposure-time equations reproduce AYO’s. Our count rates reproduce AYO’s. AYO’s own published ETC gives Earth-twin spectra at 12–18 pc that take 37–104 d. The two-month cap is stated. And yet Stark et al. (2024) Fig. 11 shows those stars reaching habitable-zone completeness of 0.5–0.8. We cannot construct a survey in which both are true. Within the four throughput parameters that could plausibly absorb the difference, over the ranges above, no setting makes them both true: Fig. 3 shows every setting we ran, and none lands on the published values.

The resolution is therefore something about how AYO counts or charges characterization within the survey that is not in the papers we have. That is Question 1 (Sec. 8).

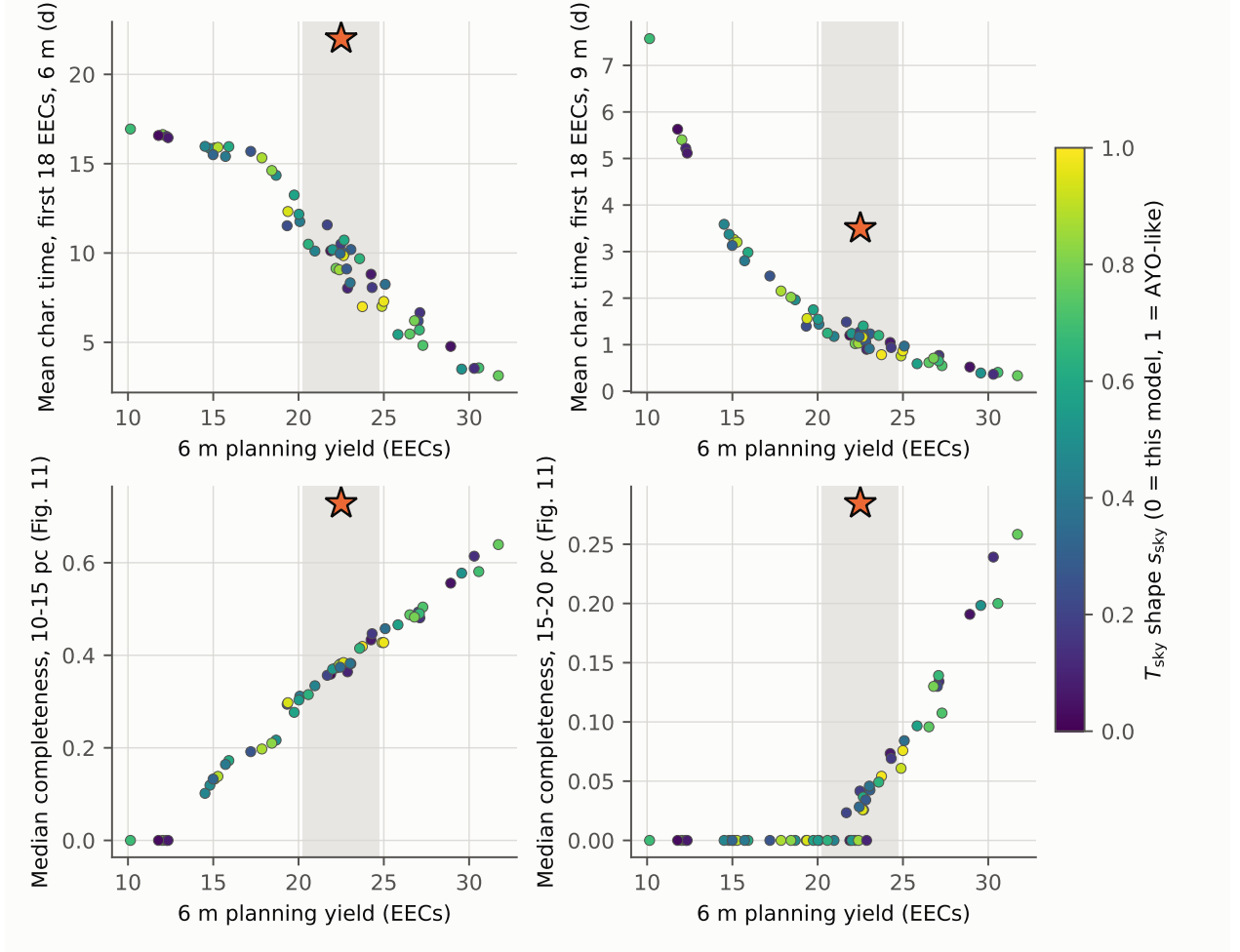


Figure 3: Every one of the 56 settings of the throughput study, plotted as a published observable against the 6 m planning yield and colored by the T_{sky} shape parameter s_{sky} . The star marks Stark et al. (2024)’s published pair of values; the shaded band is the published yield $\pm 10\%$. A setting that reproduced Stark would place a point on the star in all four panels. Instead, the points in each panel lie on one narrow curve set by the yield, with the T_{sky} shape mixed along it, and the characterization times at both apertures and the completeness at 10–15 pc miss the published values at the published yield.

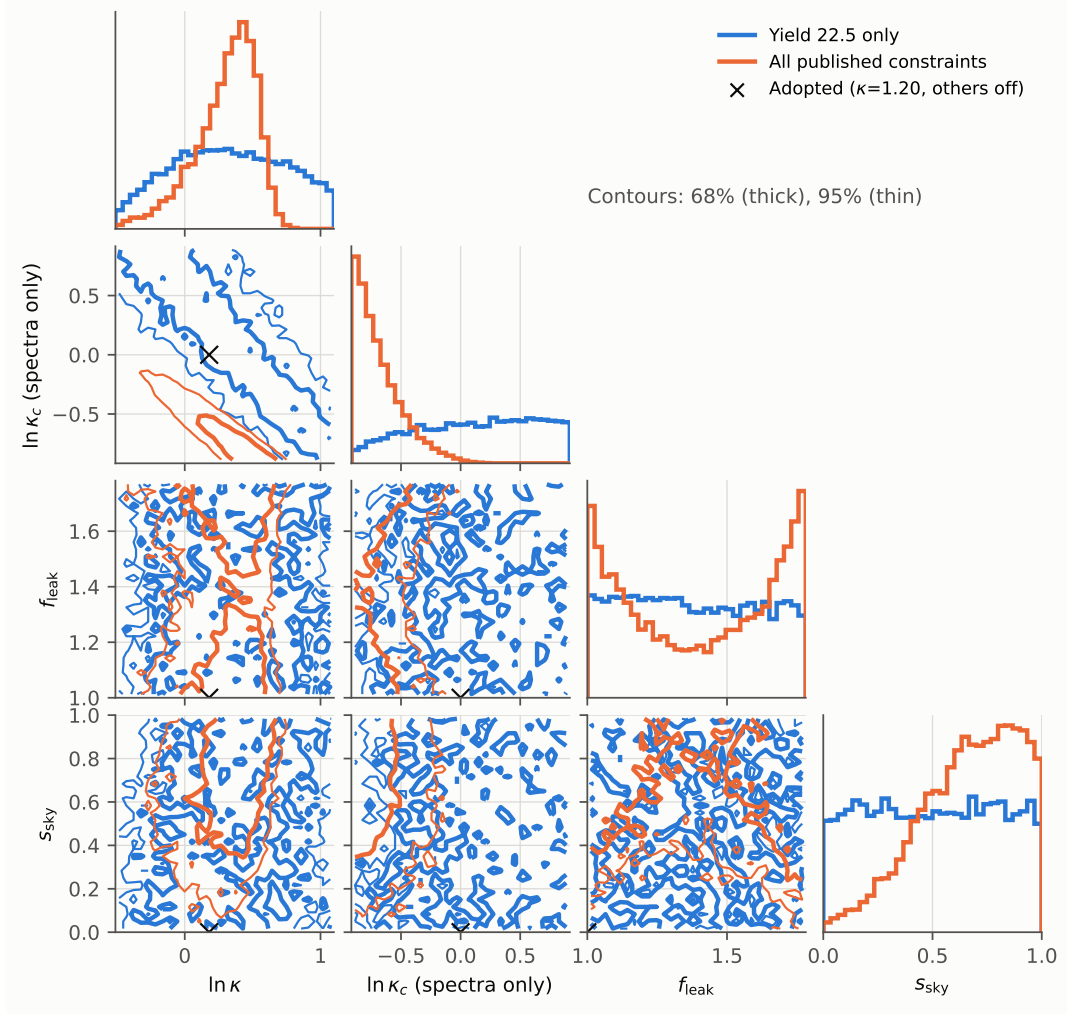


Figure 4: Posteriors over the four throughput-family parameters, from a quadratic emulator of the 56-point design whose leave-one-out error is folded into the tolerance so that emulator error cannot pose as a constraint. Blue: constrained on the published 6 m yield alone. Orange: constrained on the yield, the characterization times at 6 and 9 m, and the Fig. 11 completeness in three distance bins. The $\ln \kappa$ – $\ln \kappa_c$ panel shows the degeneracy that the published observables leave open; the orange contours run into the prior boundary on κ_c while still failing to reach the published characterization times.

6 Second discrepancy: the reading of the occurrence interval

6.1 One sentence, two readings

Stark et al. (2024) Sec. 3.6 states:

“Performing this computation for each element of our posterior, we get an $\eta_{\oplus}$ with mean and 86% confidence interval $\eta_{\oplus} = 0.26^{+0.29}_{-0.14}$.”

The underlying distribution is built by uniformly mixing the two completeness-extrapolation cases of Bryson et al. (2021), and is described as having a tail of large $\eta_{\oplus}$ values.

Propagating this requires two interpretive decisions that the sentence does not settle for a reader building a parametric law. Is 0.26 the mean, as stated, or the center of the quoted interval? And is [0.12, 0.55] an 86% interval, as stated, or the 68% that is the near-universal convention and a digit transposition away?

What we adopt. We read [0.12, 0.55] as a 68% (one-sigma) interval, not the stated 86%, and we treat its center as the *median*, not the stated mean. Concretely, $\eta_{\oplus}$ is lognormal (Eq. 4) with $z = 1$: its median is $\sqrt{0.12 \times 0.55} = 0.257$, the geometric center of the interval, its ln-width is 0.76, and its mean is 0.343, above the stated 0.26. Both choices depart from the text.

We flag this prominently because it is the kind of choice that reproductions usually bury. It was adopted because it reproduces the published Fig. 10, and a choice adopted because it reproduces the target is exactly the kind of choice that has to be justified on other grounds or declared.

6.2 A factorial decomposition

Our pipeline follows Stark’s documented albedo method, and departs from a fully text-faithful reading in one setting: the interval reading above. To test whether that departure stands alone, we pair it with the albedo method, which is the other setting a reproduction could plausibly choose differently: re-deriving each drawn-albedo planet’s exposure time instead of Stark’s per-visit flux test. The two settings form a complete 2×2 , which we evaluated end to end (Table 5).

The decomposition is unambiguous. On the total-variation distance D_{TV} to the published Fig. 10, the interval reading is worth +0.0936, the albedo method -0.0009, and the interaction +0.0006. **The interval reading accounts for 99.1% of the disagreement and the albedo method 0.9%.** The albedo method affects the agreement with Fig. 10 hardly at all; it matters only for the albedo penalty (3.6% with Stark’s test, 8.6% with re-derived exposures, against a published 12%).

Model	Peak	Mean	$P(Y < 12)$	D_{TV} to Fig. 10	Yield ratio
Adopted	12	22.4	0.26	0.037	0.2315
$\eta_{\oplus}$ as 86% (every documented choice)	15	20.6	0.19	0.131	0.2292
Albedo: exposure re-derived	11	21.0	0.28	0.038	0.2386
Albedo re-derived and 86%	15	19.5	0.22	0.132	0.2363
Constant T_{sky}	10	19.2	0.34	0.089	0.2225
86% and constant T_{sky}	12	17.4	0.30	0.146	0.2201
Stark’s own $\eta_{\oplus}$ law	7	17.8	0.41	0.155	0.2279
Stark et al. (2024)	10	21.0	0.29	—	—

Table 5: Seven models, each differing from the adopted one in named settings, evaluated end to end with every other setting, input file and random seed held fixed and nothing fitted. “Peak” is the mode of the exact Poisson mixture (Eq. 21), and D_{TV} the total-variation distance to Stark’s purple curve (Eq. 22; smaller is better). The last column is the deliverable of Sec. 7; it stays between 0.2201 and 0.2386 across the whole table.

6.3 Following every documented choice makes the agreement worse

The natural response to a reproduction that departs from its source in two places is to demand that it stop departing. We did that, and it fails (Fig. 5). With the albedo test already Stark’s, adopting the stated 86% interval as well gives a model that follows every documented choice. It moves the peak from 12 to 15 against a published 10, and the total-variation distance from 0.037 to 0.131.

This is the central empirical fact of this section, and we do not think it is a comfortable one for either side. Read one way it says our model is wrong in a manner that only the 68% reading happens to compensate. Read the other way it says the published text and the published figure are not mutually consistent, and that the figure is the better guide to what was computed. We present evidence below for the second reading, but we want to be explicit that the first cannot be excluded by anything we have.

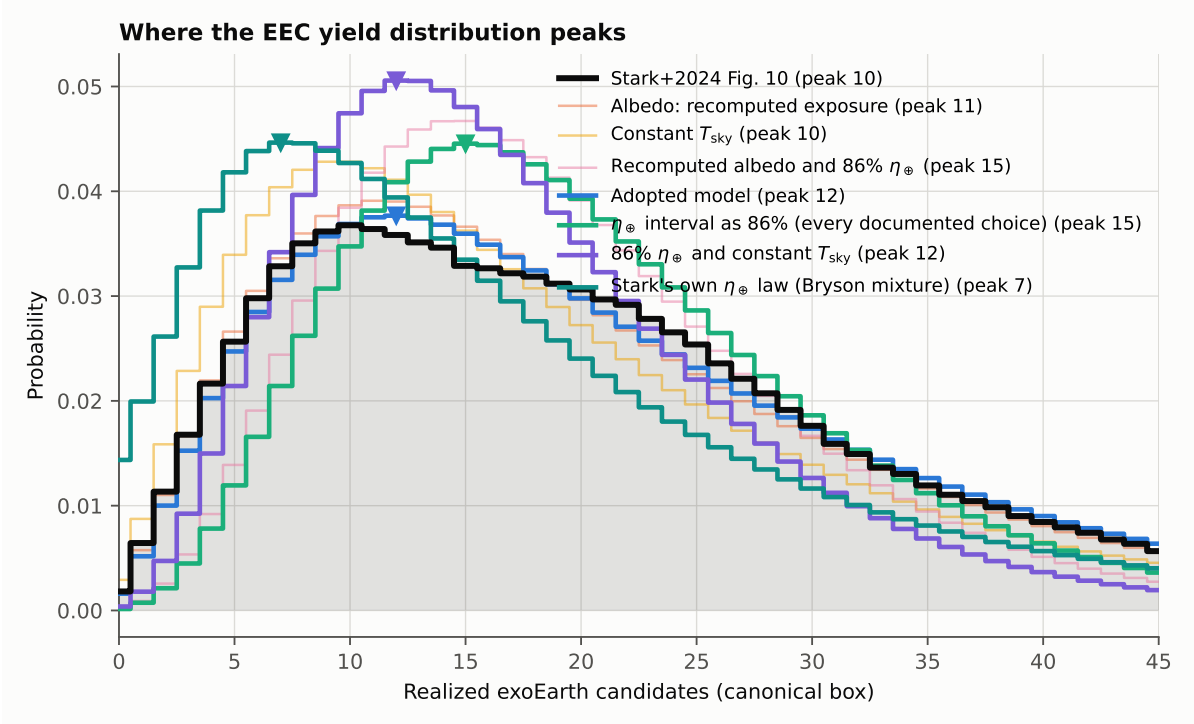


Figure 5: Realized EEC yield distributions for every model of Table 5, against the digitized Stark et al. (2024) Fig. 10 (heavy black). Each curve is the exact Poisson mixture over that model’s sampled expected yields. The one-at-a-time probes are drawn back; the combined models carry peak markers.

6.4 Testing the interval against its own source

The reading can be tested without reference to our yield model at all, by asking what the quoted interval means for the distribution it is quoted from. We took the two Bryson et al. (2021) cases, mixed them uniformly as Stark et al. (2024) describes, and rescaled the mixture by the single factor that sets its mean to the stated 0.26. No parameter is fitted.

That distribution puts 59% of its mass inside the quoted $[0.12, 0.55]$. Its own 68% interval is $[0.076, 0.429]$ and its own 86% interval is $[0.047, 0.653]$ — the latter far wider than what is printed. Its median is 0.183 when its mean is 0.26, which is strongly right-skewed and therefore inconsistent with reading 0.26 as a median, as our pipeline does.

Two conclusions follow. The quoted interval is much closer to a 68% interval of its own source than to an 86% one, which supports the transposition reading. But the mixture is also not a lognormal, and its shape cannot be represented by one: no lognormal can simultaneously carry the stated mean, the stated interval and any single confidence level. Our 68% reading may therefore not be

a corrected typo at all — it may be a width that compensates for a tail our functional form does not have. Those two explanations are observationally identical in the figure agreement, which is precisely why the agreement alone never justified the choice.

We tested the obvious remedy: replacing the lognormal with Stark’s own construction, the rescaled Bryson mixture, as the occurrence law. It does not work either. That model peaks at 7 against a published 10 and has the largest total-variation distance of any model we have built (0.155). We believe this reflects our shape transfer rather than his method: Bryson’s published distribution is for his own radius and effective-temperature selection, while Stark recomputed $\eta_{\oplus}$ from Bryson’s chains over a different one, and rescaling transfers the first moment but not the shape. The three candidate laws bracket the published curve — the Bryson mixture at 7, the 68% lognormal at 12, the 86% lognormal at 15, against 10 — and we cannot choose among them from published information. That is Question 2 (Sec. 8).

7 The counterfactual, and its robustness

We now answer the question that motivated the reimplementaion: what the baseline HWO mission would find if exoEarth candidates were defined by mass, $0.5\text{--}2\,M_{\oplus}$ with $R = M^{0.27}$, in a habitable zone of $0.96\text{--}1.20$ AU scaled as $\sqrt{L_{\star}}$.

The occurrence rate over the redefined box is 0.0520 against 0.255 over the canonical one, a ratio of 0.204. The median realized canonical yield is 18.8 (90% interval 5.6–53.4) and the median redefined yield is 4.3 (1.2–14.8). The yield ratio, with its 68% interval, is

$$\frac{Y_{\text{redefined}}}{Y_{\text{canonical}}} = 0.232^{+0.027}_{-0.021}.$$

The probability of reaching 25 candidates falls from 35.1% to 1.20%.

The yield ratio exceeds the occurrence ratio, which is at first sight surprising and is real. Two effects contribute: fewer expected detections free characterization time, which the optimizer reallocates; and truncating the outer habitable zone preferentially removes the faintest, most expensive planets, so the candidates that survive are cheaper per detection.

The result is robust to both open questions. This is the reason we are willing to report it while Secs. 5 and 6 remain open. Across the seven models of Table 5 — which span both readings of the occurrence interval, both albedo methods, the presence and absence of the T_{sky} term, and a completely different functional family for the occurrence law — the yield ratio spans 0.2201 to 0.2386, inside the 68% interval of the adopted value. The ratio is a comparison between two selections evaluated in the same model, and the systematics that trouble the absolute yield largely divide out of it.

8 Discussion: the two questions

We have described a reproduction that succeeds on most of the published record and fails on two points. We now argue that those two points are not a list of remaining work but the hinge of the whole exercise, and that each reduces to a question of fact that one email can settle.

Question 1. How does the AYO baseline survey characterize its 10–20 pc targets within the two-month limit?

Stark et al. (2024) Fig. 11 shows habitable-zone completeness of 0.5–0.8 for targets at 10–20 pc. AYO’s own exposure-time calculator (Stark et al. 2025) gives Earth-twin spectra at those distances of 37–104 d. The characterization limit is two months. We reproduce the ETC to 0.94–1.02 on

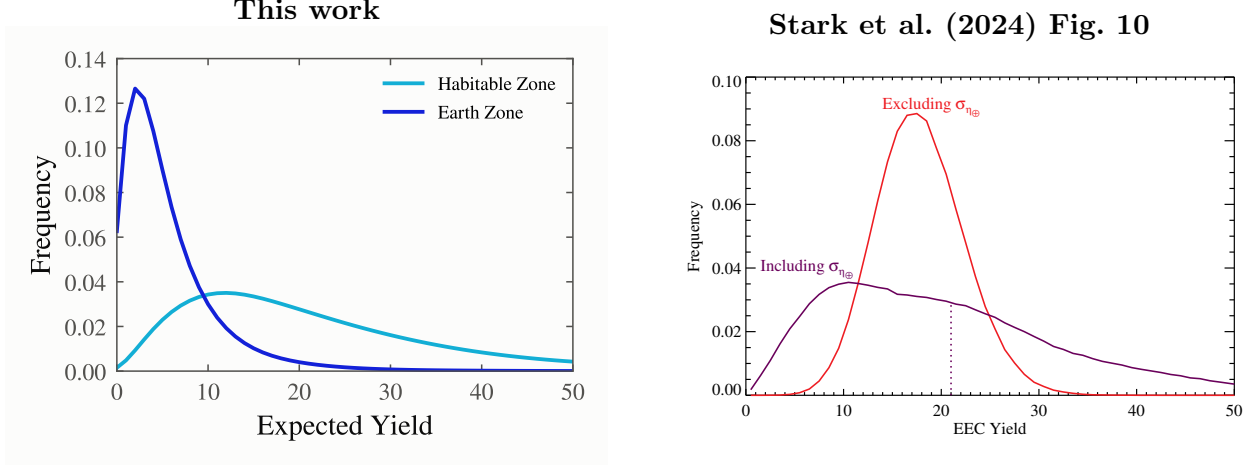


Figure 6: Left: EEC yield distributions for the canonical habitable-zone selection (pale blue) and for the narrowed, mass-selected Earth zone (dark blue), marginalized over the occurrence posterior and the exozodi draws and including exoplanet sampling uncertainty. Each curve is the exact Poisson mixture over the sampled expected yields. Right: the published distributions for the canonical selection, copied unaltered from the arXiv source. Our pale blue curve is the counterpart of Stark’s purple curve (“Including $\sigma_{\eta_{\oplus}}$ ”), not his red one: both describe the same selection, the same 6m design and the same sources of uncertainty (the spread in $\eta_{\oplus}$, exozodi, albedo and planet sampling). They should share a broad peak near 10, a mean near 21 and a long tail toward high yields; ours peaks at 12 (published 10), with mean 22.4 (21.0), median 19 (18), $P(Y < 12) = 0.26$ (0.29), and a total-variation distance of 0.037 between the two curves. Stark’s red curve holds $\eta_{\oplus}$ fixed (“Excluding $\sigma_{\eta_{\oplus}}$ ”), which removes most of the spread; its counterpart in this model is the fixed- $\eta_{\oplus}$ distribution, with mean 17.73 against his 17.35, and it is not drawn here. The Earth zone has no published counterpart. The yield range follows Stark’s Fig. 10; our abscissa is labeled in plain English rather than with his “EEC yield”.

AYO’s own inputs, and across the four throughput parameters that could absorb the difference, including a T_{sky} shaped like AYO’s, no setting we ran does (Sec. 5, Fig. 3). Either those stars are characterized under conditions more favorable than an Earth twin at maximum elongation, or the completeness plotted in Fig. 11 is not gated on characterization, or characterization is charged to the budget differently from the way we charge it, or AYO’s DMVC6 T_{sky} map departs from ours by more than its vortex benchmark suggests (Sec. 3.5). Any of these would be a one-sentence answer, and each implies a different correction.

Question 2. In Sec. 3.6, is $\eta_{\oplus} = 0.26$ the mean or the median, and is $[0.12, 0.55]$ an 86% or a 68% interval?

The text says mean and 86%. Our pipeline uses median and 68%, and that combination reproduces the published Fig. 10 while the stated combination does not (Sec. 6). The quoted interval contains 59% of the mass of its own source distribution, which is much nearer 68% than 86%. “86” is a digit transposition of “68”, and the phrase occurs exactly once in the paper, which otherwise quotes standard deviations throughout. We are not in a position to assert a typographical error in someone else’s paper on that evidence, and we do not. But the question is answerable immediately by the authors, and nothing else we can do resolves it.

Why the reproduction hinges on these

It would be possible to read the preceding sections as a report of two modest residuals in an otherwise successful reproduction. We think that reading is wrong, for three reasons.

First, they control the quantities that matter for mission design. The characterization time budget determines the target mix, and the target mix determines which stars must be surveyed for exozodi

in advance, which drives a precursor-mission argument. The occurrence interval controls the width of the yield distribution, and the width determines $P(Y \geq 25)$, which is the quantity the decadal recommendation is written in. A reproduction that matches the mean yield but not the width does not reproduce the statement the mission is designed against.

Second, the agreement we do have is partly contingent on the choices in question. Our agreement with the shape of Fig. 10 holds under the 68% reading and fails under the 86% one. We cannot simultaneously claim that our model is validated by that agreement and that the reading which produces it is an open question. Honesty requires reporting the agreement as conditional, and Question 2 is the condition.

Third, the two are not independent in their consequences. Both act on the same downstream quantity. The characterization-time discrepancy moves the target mix and hence the level; the occurrence reading moves the width. A reader trying to judge whether our model is fit for a given purpose needs both answered, because a model with the right level and the wrong width and a model with the wrong level and the right width fail different applications.

We therefore present this work not as a completed reproduction with caveats, but as a reproduction that has been carried as far as the published record permits, together with the two specific facts that would carry it the rest of the way.

What we are not claiming

We are not claiming an error in AYO. Every discrepancy reported here is equally consistent with an error in our reimplementation that we have failed to find, and we have found and fixed seventeen such errors already (Appendix C); there is no reason to believe the count is final. What we do claim is narrower and, we think, defensible: that the published description is not sufficient to determine these two quantities, and that we have exhausted the interpretations available to us.

9 Conclusions

1. An independent reimplementation of AYO, built from the published equations and parameters with no parameter fitted to the published yield, reproduces the Stark et al. (2024) baseline 6 m exoEarth candidate yield to 7.0% and agrees with 21 of 24 independent published comparisons.
2. The realized-yield distribution reproduces the shape of the published one (total-variation distance 0.037). The peak should be computed as an exact Poisson mixture; the mode of a histogram of realized draws is too noisy to support the comparison.
3. Stark’s own per-visit albedo test, adopted here, gives an albedo penalty of 3.6% against the 12% he reports.
4. Spectral characterization times remain too steep in target difficulty and aperture. A 56-point factorial design establishes that no member of the throughput-parameter family reaches the published characterization times at the published yield.
5. The $\eta_{\oplus}$ interval of Sec. 3.6 admits two readings that differ materially. A 2×2 decomposition attributes 99.1% of the residual disagreement with the published yield distribution to that single choice, and following every documented choice simultaneously makes the agreement worse rather than better.
6. Both residues reduce to questions of fact that the AYO authors can answer in a sentence, and we state them explicitly in Sec. 8.

7. For the counterfactual that motivated the work — a mass-selected sample in a narrowed habitable zone — the yield ratio is $0.232^{+0.027}_{-0.021}$ (68% interval), and it stays between 0.2201 and 0.2386 across every model variant spanning both open questions.

A Verification against the published record

Check	Published	Model	Diff.	Tol.	Verdict
eta.Earth over the canonical box (SAG13)	0.24	0.2404	0%	3%	agrees
Uncalibrated 6 m yield	22.5	20.92	7%	20%	agrees
Throughput factor needed to reach 22.5	1	1.195	20%	100%	agrees
Sampling distribution mean (Fig. 4)	22.5	20.91	7%	20%	agrees
Sampling distribution sigma (Fig. 4)	5	4.567	9%	25%	agrees
Albedo penalty on expected yield	0.12	0.03608	70%	50%	disagrees
P25 including sigma_eta, 6 m	0.32	0.3514	10%	25%	agrees
Yield-aperture exponent	1.9	1.813	5%	15%	agrees
Exozodi-sampling penalty on expected yield	0.111	0.1217	10%	50%	agrees
P25 excluding sigma_eta, 6 m	0.06	0.06264	4%	50%	agrees
Mean characterization time, first 18 EECs, 6 m (days)	22	14.34	35%	30%	disagrees
Mean characterization time, first 18 EECs, 9 m (days)	3.5	2.053	41%	30%	disagrees
Characterization-time ratio 6 m / 9 m	6.2	6.987	13%	20%	agrees
P25 including sigma_eta at 6 m	0.32	0.3508	10%	25%	agrees
P25 including sigma_eta at 7 m	0.53	0.5271	1%	25%	agrees
P25 including sigma_eta at 8 m	0.67	0.6573	2%	25%	agrees
P25 including sigma_eta at 9 m	0.78	0.755	3%	25%	agrees
Fixed-eta yield mean (Fig. 10 red)	17.35	17.73	2%	10%	agrees
Realized-yield distribution median	18	19	6%	15%	agrees
Realized-yield fraction below 12 EECs	0.2907	0.2597	11%	35%	agrees
Realized-yield distribution mean	21	22.45	7%	30%	agrees
Realized-yield distribution mode	10	11	10%	25%	agrees
Maximum luminosity among reachable targets	20	12.5	38%	60%	agrees
Maximum distance among reachable targets	25	22.02	12%	60%	agrees

Table 6: The 24 independent comparisons of step A09, of which 21 agree. Published values taken from Fig. 10 are digitized from the figure’s vector content stream. Comparisons against quantities that entered the model as inputs are tagged as tuned and excluded from the count.

B Published figures beside their model counterparts

C Ledger of hypotheses tested

The reproduction proceeded by forming hypotheses about the source of each disagreement and testing them. We record the full ledger because the distinction between a hypothesis rejected on evidence and a real defect that turned out not to explain the symptom is load-bearing: a defect can be genuine, worth fixing, and irrelevant to the symptom that led to it. Four separate times an apparent agreement rested on two errors that cancelled, which is the reason this paper reports conditional agreement rather than agreement.

D Every contributing star

Star	SpT	d (pc)	L ($L_{\odot}$)	Earth zone (mas)	classic HZ (mas)	C_{HZ}	C_{EZ}	N_{HZ}	N_{EZ}	ratio
alf Cen B	K1V	1.35	0.552	529–661	524–921	0.49	0.65	0.250	0.052	0.21
HD 95735	M2+V	2.55	0.0239	58–73	58–101	0.99	1.00	0.249	0.052	0.21

Star	SpT	d (pc)	L ($L_{\odot}$)	Earth zone (mas)	classic HZ (mas)	C_{HZ}	C_{EZ}	N_{HZ}	N_{EZ}	ratio
tau Cet	G8V	3.65	0.52	189–237	188–330	0.98	1.00	0.248	0.052	0.21
eps Ind	K5V	3.64	0.222	124–155	123–216	0.98	1.00	0.247	0.052	0.21
61 Cyg A	K5V	3.50	0.158	109–136	108–190	0.98	1.00	0.247	0.052	0.21
61 Cyg B	K7V	3.50	0.104	89–111	88–154	0.98	1.00	0.246	0.052	0.21
HD 217987	M2V	3.29	0.0413	59–74	59–103	0.97	1.00	0.245	0.052	0.21
V* AX Mic	M1V	3.97	0.105	78–98	77–136	0.96	1.00	0.239	0.051	0.21
70 Oph A	K0V	5.11	0.514	135–168	133–234	0.95	1.00	0.237	0.051	0.22
del Equ	F7(V)+G	4.51	0.265	110–137	108–191	0.95	1.00	0.236	0.052	0.22
omi02 Eri	K0V	5.01	0.428	125–157	124–218	0.95	1.00	0.236	0.052	0.22
eps Eri	K2V	3.22	0.342	174–218	173–303	0.96	1.00	0.236	0.051	0.22
70 Oph B	K4V	5.10	0.162	76–95	75–132	0.92	0.99	0.229	0.051	0.22
e Eri	G6V	6.04	0.663	129–162	128–225	0.91	0.99	0.223	0.051	0.23
36 Oph	K2V+K1V	5.95	0.334	93–117	92–162	0.90	0.97	0.222	0.050	0.22
sig Dra	K0V	5.76	0.433	110–137	108–191	0.89	0.98	0.221	0.051	0.23
IRAS 20079-3614		6.01	0.26	81–102	81–142	0.88	0.94	0.221	0.049	0.22
HD 88230	K6VeFe-	4.87	0.117	68–84	67–118	0.89	0.97	0.220	0.050	0.23
HD 131977	K4V	5.89	0.223	77–96	76–134	0.87	0.94	0.218	0.049	0.22
V* GX And	M2V	3.56	0.0239	42–52	41–72	0.86	0.88	0.217	0.046	0.21
36 Oph B	K1V	5.95	0.331	93–116	92–162	0.87	0.96	0.216	0.049	0.23
eta Cas	F9V	5.92	1.27	182–228	181–317	0.88	0.99	0.214	0.051	0.24
ksi Boo	G7Ve+K5	6.75	0.554	106–132	105–184	0.86	0.92	0.213	0.048	0.22
HD 219134	K3V	6.54	0.29	79–99	78–137	0.84	0.89	0.211	0.046	0.22
del Pav	G8IV	6.10	1.25	176–220	174–306	0.87	0.99	0.210	0.050	0.24
V* V2215 Oph	K5V	5.95	0.158	64–80	63–111	0.84	0.87	0.210	0.045	0.22
107 Psc	K1V	7.64	0.458	85–106	84–148	0.83	0.87	0.205	0.045	0.22
mu. Cas	G5Vb	7.68	0.479	87–108	86–151	0.79	0.82	0.194	0.042	0.22
HD 173739	M3V	3.52	0.016	35–43	34–60	0.76	0.69	0.193	0.036	0.19
61 Vir	G6.5V	8.53	0.841	103–129	102–179	0.78	0.84	0.192	0.043	0.22
HD 16160	K3V	7.23	0.282	71–88	70–123	0.77	0.80	0.191	0.041	0.22
bet CVn	G0V	8.47	1.22	125–156	124–218	0.78	0.85	0.190	0.043	0.23
HD 156384	K3V+K5V	6.97	0.192	60–76	60–105	0.74	0.77	0.186	0.040	0.21
HD 4628	K2.5V	7.43	0.3	71–88	70–123	0.73	0.75	0.181	0.039	0.22
V* TW PsA	K4Ve	7.60	0.196	56–70	55–97	0.72	0.78	0.180	0.041	0.23
p Eri A	K2V	8.19	0.345	69–86	68–120	0.72	0.77	0.179	0.040	0.22
chi01 Ori	G0V	8.70	1.11	116–145	115–202	0.73	0.79	0.177	0.041	0.23
p Eri B	K2V	8.20	0.305	65–81	64–112	0.71	0.76	0.177	0.039	0.22
ksi Boo B	K5Ve	6.75	0.127	51–63	50–88	0.70	0.74	0.177	0.038	0.22
HD 192310	K2+V	8.81	0.405	69–87	69–121	0.71	0.79	0.175	0.041	0.23
ksi UMa	F8.5:V+	8.73	0.796	98–123	97–171	0.71	0.77	0.175	0.039	0.22
ksi UMa A	F8.5:V	8.73	1.26	123–154	122–214	0.72	0.80	0.173	0.040	0.23
zet Tuc	F9.5V	8.61	1.28	126–158	125–220	0.70	0.76	0.171	0.039	0.23
41 Ara	G9V	8.79	0.461	74–93	73–129	0.69	0.75	0.169	0.039	0.23
HD 131976	M1.5V	5.90	0.0723	44–55	43–76	0.67	0.67	0.169	0.035	0.21
kap01 Cet	G5V	9.27	0.881	97–122	96–169	0.69	0.76	0.168	0.039	0.23
HD 102365	G2V	9.32	0.849	95–119	94–165	0.68	0.77	0.166	0.040	0.24
bet Com	F9.5V	9.20	1.42	124–155	123–216	0.69	0.76	0.165	0.039	0.23
alf Cen A	G2V	1.35	1.66	918–1147	908–1596	0.18	0.24	0.164	0.030	0.18
61 UMa	G8V	9.57	0.625	79–99	78–138	0.67	0.76	0.163	0.039	0.24
chi Dra	F7V	8.30	2.19	171–214	170–298	0.69	0.83	0.163	0.041	0.25
gam Pav	F9VFe-1	9.26	1.46	125–157	124–218	0.67	0.76	0.160	0.039	0.24
pi.03 Ori	F6V	8.02	2.85	202–253	200–352	0.68	0.86	0.157	0.043	0.27
mu.01 Her	G5IV	8.33	2.68	188–235	186–328	0.67	0.86	0.156	0.042	0.27
alf Men	G7V	10.21	0.866	87–109	87–152	0.64	0.74	0.154	0.038	0.25
gam Lep	F6V	8.90	2.36	166–207	164–288	0.66	0.79	0.154	0.040	0.26
bet Hyi	G0V	7.48	3.57	243–303	240–422	0.64	0.92	0.152	0.044	0.29
del Tri	G0.5VFe	10.91	1.21	97–121	96–168	0.63	0.80	0.152	0.041	0.27
HD 10780	K0V	10.04	0.536	70–88	69–122	0.60	0.73	0.147	0.038	0.26
HD 32147	K3+V	8.84	0.29	58–73	58–102	0.59	0.67	0.146	0.035	0.24
HD 79210	K7V	6.33	0.0813	43–54	43–75	0.57	0.52	0.144	0.027	0.19
tet Per	F8V	11.14	2.32	131–164	130–228	0.61	0.82	0.142	0.041	0.29
12 Oph	K1V	9.89	0.46	66–82	65–114	0.58	0.70	0.142	0.036	0.26
iot Per	G0V	10.57	2.26	136–171	135–237	0.61	0.78	0.141	0.039	0.28
V* AK Lep	K3	8.89	0.279	57–71	56–99	0.56	0.63	0.139	0.033	0.24
del Eri	K0+IV	9.09	3.29	192–240	190–333	0.59	0.82	0.136	0.040	0.29
HD 36395	M1.5Ve	5.70	0.0594	41–51	41–71	0.53	0.47	0.134	0.024	0.18
11 LMi	G8Va	11.23	0.801	77–96	76–133	0.53	0.70	0.128	0.036	0.28
zet Dor	F9VFe-0	11.69	1.48	100–125	99–174	0.53	0.72	0.125	0.037	0.29
lam Ser	G0-V	11.90	2.03	115–144	114–200	0.54	0.73	0.125	0.037	0.30
zet TrA	F9V	12.07	1.34	92–115	91–160	0.51	0.71	0.121	0.036	0.30
20 Crt	K0V	9.56	0.371	61–77	61–106	0.48	0.57	0.120	0.029	0.24
lam Aur	G1.5IV-	12.56	1.78	102–128	101–178	0.50	0.72	0.117	0.036	0.31
gam Ser	F6V	11.16	2.96	148–185	147–258	0.50	0.71	0.114	0.034	0.30
HD 50281	K3.5V	8.74	0.219	51–64	51–89	0.46	0.41	0.114	0.021	0.19
HD 42581	M1V	5.76	0.0515	38–47	37–66	0.44	0.19	0.111	0.010	0.09

Star	SpT	d (pc)	L ($L_{\odot}$)	Earth zone (mas)	classic HZ (mas)	C_{HZ}	C_{EZ}	N_{HZ}	N_{EZ}	ratio
HD 79211	M0V	6.33	0.0707	40–50	40–70	0.44	0.24	0.111	0.013	0.11
zet02 Ret	G1V	12.04	1.02	80–100	80–140	0.46	0.63	0.110	0.032	0.29
bet Vir	F9V	11.00	3.69	168–209	166–292	0.48	0.74	0.108	0.035	0.33
iot Peg	F5V	11.90	3.57	152–191	151–265	0.48	0.75	0.108	0.035	0.33
36 UMa	F8V	12.94	1.63	95–118	94–165	0.46	0.66	0.107	0.034	0.31
zet01 Ret	G2.5VHd	12.04	0.795	71–89	70–124	0.44	0.53	0.106	0.027	0.26
HD 17925	K1V	10.36	0.401	59–73	58–102	0.41	0.38	0.100	0.019	0.19
HD 191849	M0V	6.16	0.0616	39–48	38–67	0.38	0.15	0.095	0.008	0.08
54 Psc	K0.5V	11.10	0.529	63–79	62–109	0.38	0.39	0.092	0.020	0.22
ups And	F9V	13.47	3.38	131–164	130–228	0.41	0.67	0.090	0.032	0.35
HD 157881	K7V	7.71	0.123	44–55	43–76	0.35	0.16	0.087	0.008	0.09
HD 103095	K1V_Fe-	9.17	0.235	51–63	50–88	0.34	0.23	0.085	0.012	0.14
10 Tau	F9IV-V	13.92	3.16	123–153	121–213	0.37	0.61	0.084	0.029	0.35
47 UMa	G1-VFe-	13.88	1.6	87–109	86–152	0.36	0.50	0.084	0.025	0.30
gam Vir	F1-F2V	12.02	4.55	170–213	169–296	0.38	0.65	0.083	0.029	0.36
alf CMi	F5IV-V+	3.50	7.21	737–921	729–1282	0.15	0.30	0.080	0.028	0.36
tau01 Eri	F7V	14.27	2.67	110–137	109–191	0.34	0.53	0.080	0.026	0.33
iot Psc	F7V	13.66	3.39	129–162	128–225	0.35	0.59	0.079	0.028	0.36
HD 119850	M2V	5.43	0.0397	35–44	35–61	0.30	0.06	0.076	0.003	0.04
alf For	F6V	14.00	4.28	142–177	140–247	0.35	0.60	0.076	0.027	0.36
HD 104304	G8IV	12.76	0.892	71–89	70–124	0.32	0.34	0.076	0.017	0.23
HD 223778	K3V	10.89	0.399	56–70	55–97	0.30	0.19	0.074	0.010	0.13
eta Cas B	K7Ve	5.93	0.0513	37–46	36–64	0.29	0.04	0.074	0.002	0.03
alf Com	F5V+F6V	14.32	3	116–145	115–202	0.32	0.51	0.072	0.025	0.35
HD 147513	G5V	12.89	0.998	74–93	74–129	0.29	0.35	0.071	0.018	0.25
V* DE Boo	K0.5V	11.63	0.541	61–76	60–106	0.29	0.21	0.070	0.011	0.15
HD 13445	K1.5V	10.76	0.406	57–71	56–99	0.28	0.20	0.070	0.010	0.14
gam Vir B	F0mF2V	12.74	5.27	173–216	171–301	0.32	0.56	0.069	0.025	0.37
HD 122064	K3V	10.06	0.291	51–64	51–89	0.28	0.12	0.068	0.006	0.09
58 Eri	G2.5IV-	13.23	0.966	71–89	71–124	0.28	0.26	0.066	0.013	0.20
psi Cap	F5V	14.63	3.72	127–158	125–220	0.29	0.48	0.064	0.023	0.35
85 Peg	G5VbFe-	12.32	0.674	64–80	63–111	0.26	0.21	0.064	0.011	0.17
111 Tau	F8V	14.57	1.78	88–110	87–153	0.27	0.35	0.062	0.017	0.28
HD 10307	G1V	14.67	1.89	90–112	89–157	0.26	0.37	0.061	0.019	0.31
26 Dra	F9V+K3V	14.43	1.48	81–101	80–141	0.26	0.31	0.060	0.016	0.26
HD 211415	G0V	14.06	1.17	74–92	73–129	0.25	0.23	0.059	0.012	0.20
chi Her	F8VFe-2	15.90	3.16	107–134	106–187	0.26	0.40	0.058	0.020	0.34
zet Her	G0IV	10.80	7.26	240–300	237–417	0.26	0.49	0.055	0.020	0.36
nu. Phe	F9VFe+0	15.25	2.03	90–112	89–156	0.24	0.30	0.055	0.015	0.28
b Aql	G7IVHde	14.91	1.7	84–105	83–146	0.23	0.28	0.053	0.014	0.27
alf Crv	F1V	14.97	4.34	134–167	132–232	0.23	0.39	0.051	0.018	0.35
HD 84117	F9V	14.95	2.01	91–114	90–159	0.22	0.30	0.051	0.015	0.29
h Dra	F8V	15.33	2.28	95–118	94–164	0.22	0.29	0.051	0.015	0.29
tau Boo	F7IV-V	15.60	3.18	110–137	109–191	0.23	0.33	0.050	0.016	0.32
sig Boo	F4VkF2m	15.75	3.22	109–137	108–190	0.21	0.29	0.047	0.014	0.30
eta Lep	F2V	14.95	5.66	153–191	151–266	0.22	0.37	0.047	0.016	0.35
171 Pup	F9V	15.27	1.57	79–99	78–137	0.19	0.19	0.046	0.010	0.21
m Tau	G4V	15.91	2.41	94–117	93–163	0.20	0.27	0.045	0.013	0.29
18 Sco	G2Va	14.13	1.1	71–89	70–124	0.19	0.14	0.045	0.007	0.16
mu. Ara	G3IV-V	15.60	1.9	85–106	84–148	0.20	0.22	0.044	0.011	0.25
20 LMi	G3VaHde	14.92	1.37	75–94	75–131	0.18	0.15	0.043	0.008	0.18
b Her	F6V	15.76	2.02	87–108	86–151	0.19	0.21	0.043	0.010	0.24
HD 176051	F9V+K1V	14.89	1.42	77–96	76–134	0.18	0.16	0.043	0.008	0.19
HD 69830	G8:V	12.58	0.609	60–74	59–104	0.18	0.05	0.043	0.002	0.06
HD 160346	K3-V	10.71	0.32	51–63	50–88	0.17	0.02	0.042	0.001	0.03
ksi Peg	F6V	16.39	4.76	128–160	126–222	0.19	0.30	0.042	0.013	0.32
rho01 Cnc	K0IV-V	12.58	0.635	61–76	60–106	0.17	0.06	0.041	0.003	0.07
HD 166620	K2V	11.09	0.363	52–65	52–91	0.16	0.02	0.040	0.001	0.03
tet UMa	F7V	13.53	7.9	199–249	197–347	0.18	0.35	0.037	0.013	0.36
eta Boo	G0IV	11.34	8.79	251–314	248–437	0.16	0.33	0.035	0.013	0.37
bet TrA	F1V	12.42	8.71	228–285	226–397	0.16	0.30	0.033	0.012	0.35
I Car	F3V	16.28	5.25	135–169	134–235	0.14	0.22	0.031	0.010	0.33
HD 172051	G6V	13.02	0.682	61–76	60–106	0.12	0.02	0.029	0.001	0.03
i Boo	G0Vn	12.95	0.639	59–74	59–103	0.12	0.01	0.028	0.000	0.02
gam CrA A	F8V	17.03	2.47	89–111	88–154	0.13	0.12	0.028	0.006	0.20
tau01 Hya	F5V	18.24	4.05	106–132	105–184	0.13	0.19	0.028	0.008	0.30
HD 3443	K1V+G	15.53	1.32	71–89	70–123	0.12	0.05	0.027	0.002	0.08
9 Pup	G0V	16.67	1.95	80–101	80–140	0.12	0.07	0.026	0.004	0.14
HD 188088	K3VaCN1	14.14	0.803	61–76	60–106	0.11	0.01	0.026	0.000	0.02
nu.02 Lup	G3/5V	14.74	1.04	66–83	66–115	0.11	0.02	0.026	0.001	0.03
pi.01 UMa	G0.5V	14.43	0.972	66–82	65–114	0.11	0.01	0.026	0.001	0.03
phi02 Cet	F7V	15.92	1.79	81–101	80–140	0.11	0.07	0.025	0.004	0.15
gam CrA	F8V+F8V	17.89	4.8	118–147	116–204	0.11	0.13	0.023	0.006	0.26
HD 37394	K1	12.26	0.477	54–68	53–94	0.09	0.00	0.023	0.000	0.00

Star	SpT	d (pc)	L ($L_{\odot}$)	Earth zone (mas)	classic HZ (mas)	C_{HZ}	C_{EZ}	N_{HZ}	N_{EZ}	ratio
HD 115404	K2V	10.98	0.302	48–60	48–84	0.09	0.00	0.022	0.000	0.00
del Aql	F1IIV-V(	15.37	8.26	179–224	178–312	0.11	0.20	0.022	0.008	0.35
70 Vir	G4Va	18.09	3.1	93–117	93–163	0.10	0.10	0.022	0.005	0.22
51 Peg	G2IV	15.51	1.38	73–91	72–126	0.09	0.03	0.021	0.002	0.07
g Lup	F3/5V	17.39	3.33	101–126	100–175	0.10	0.09	0.021	0.004	0.20
58 Oph	F5V	17.17	2.72	92–115	91–160	0.09	0.06	0.020	0.003	0.15
HD 74576	K3V	11.19	0.318	48–60	48–84	0.08	0.00	0.020	0.000	0.00
tau06 Eri	F5IV-V	17.78	5.15	123–153	121–213	0.09	0.11	0.020	0.005	0.26
psi05 Aur	F9V	16.61	1.82	78–97	77–136	0.08	0.02	0.019	0.001	0.06
ksi Oph	F2V	17.50	4.35	114–143	113–199	0.09	0.09	0.018	0.004	0.22
HD 103932	K4+V	10.17	0.212	43–54	43–76	0.07	0.00	0.018	0.000	0.00
HD 217357	K7+Vk	8.23	0.101	37–46	37–64	0.06	0.00	0.015	0.000	0.00
rho Gem	F1V	17.83	5.44	126–157	124–218	0.07	0.08	0.015	0.003	0.22
HD 5015	F9V	18.90	3.62	97–121	96–168	0.07	0.04	0.015	0.002	0.14
HD 207129	G2V	15.55	1.21	68–85	67–118	0.06	0.00	0.014	0.000	0.01
tet Cyg	F3+V	18.42	4.27	108–135	107–187	0.07	0.05	0.014	0.002	0.17
HD 114837	F6VFe-0	18.23	2.99	91–114	90–158	0.06	0.03	0.013	0.002	0.12
eta Crv	F2V	18.24	4.77	115–144	114–200	0.06	0.05	0.012	0.002	0.17
10 UMa	F3V+G5V	18.79	6.73	132–166	131–230	0.06	0.07	0.012	0.003	0.23
HD 32450	K7V	8.36	0.11	38–48	38–66	0.05	0.00	0.012	0.000	0.00
e Vir	G0V	17.57	2.19	81–101	80–141	0.05	0.01	0.010	0.000	0.03
HD 72673	K1V	12.16	0.403	50–63	50–87	0.04	0.00	0.010	0.000	0.00

Stars with an expected yield of at least 0.01 EEC in either zone’s survey, in order of their classic-zone yield. Zone edges are angular separations in milliarcseconds; C is the completeness reached, averaged over exozodi draws; N is the expected number of candidates. The complete table is `TabulateEarthZones/earthZoneTable.csv`.

References

- [bryson2021] Bryson, S., et al. 2021, *AJ*, 161, 36 (arXiv:2010.14812).
- [savransky2016] Savransky, D., & Garrett, D. 2016, *JATIS*, 2, 011006 (EXOSIMS).
- [stark2014] Stark, C. C., Roberge, A., Mandell, A., & Robinson, T. D. 2014, *ApJ*, 795, 122.
- [stark2019] Stark, C. C., et al. 2019, *JATIS*, 5, 024009 (arXiv:1904.11988).
- [stark2024] Stark, C. C., et al. 2024, *JATIS* (arXiv:2405.19418).
- [stark2025] Stark, C. C., et al. 2025, arXiv:2502.18556 (ETC benchmark).
- [tuchow2024] Tuchow, N. W., et al. 2024, the HWO Preliminary Input Catalog (HPIC).

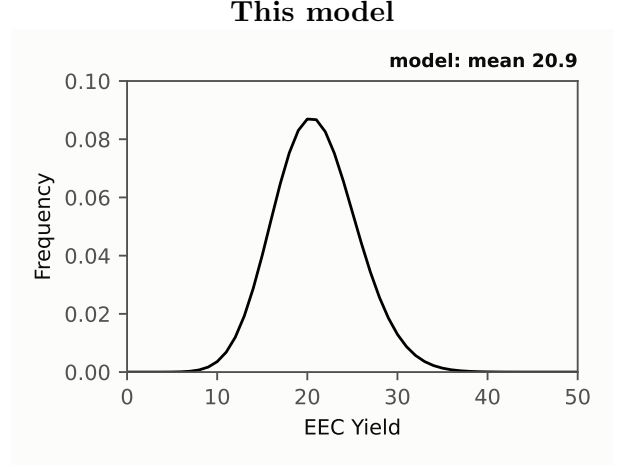
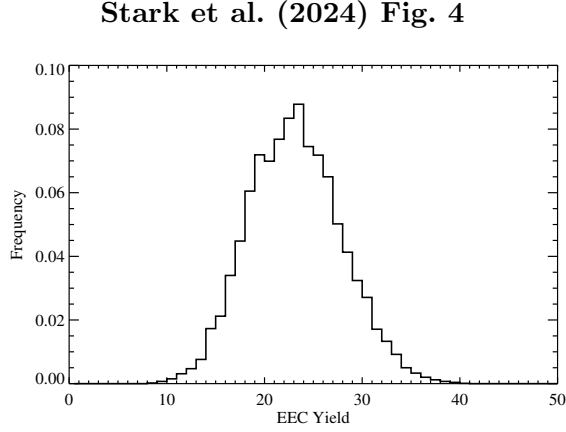


Figure 7: Planet sampling as the only source of uncertainty: $A_G = 0.2$, exozodi fixed at 3 zodis, and Poisson counting of planets. Published: mean 22.5, standard deviation 21% of the mean. Model: mean 20.9, with nothing fitted to the published yield.

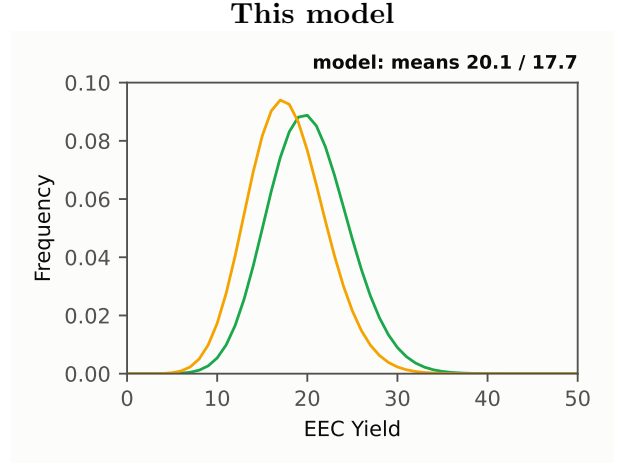
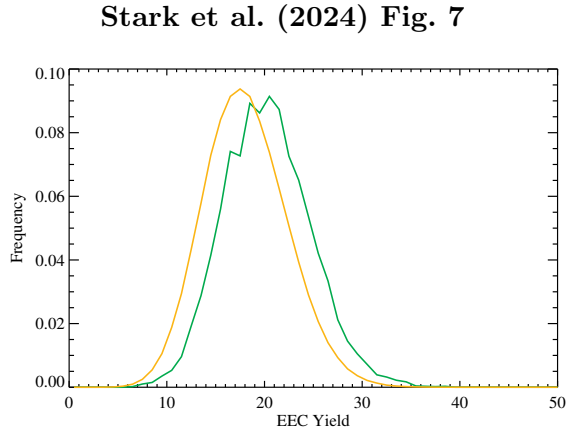


Figure 8: Yield with albedo drawn and exozodi fixed (green), and with exozodi drawn as well (orange), both at fixed $\eta_\oplus$. Published means: 19.8 and 17.6. Model means: 20.1 and 17.7, over 200 draws.

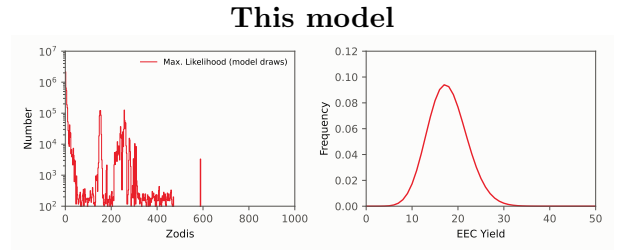
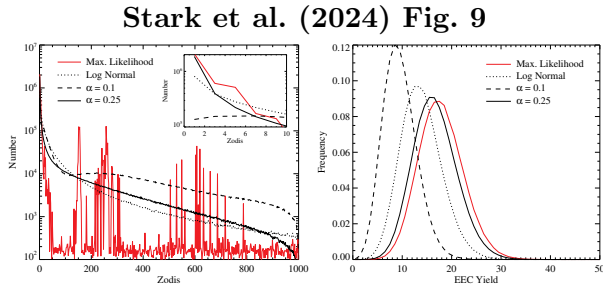
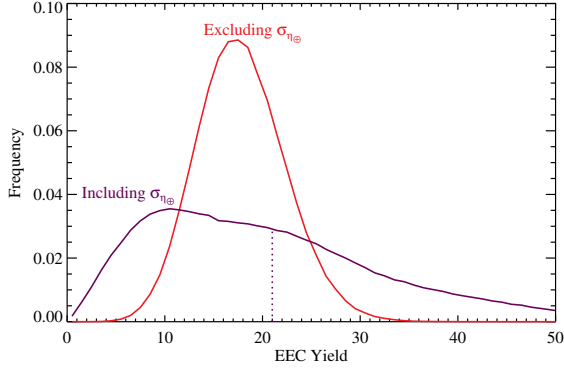


Figure 9: Left: exozodi levels drawn from the LBTI HOSTS maximum-likelihood distribution (red). The model panel is a histogram of the levels the pipeline actually drew, scaled to Stark's 500×10^4 draws; its median is 3.00 zodis. Stark's other three distributions are not modeled. Right: the yield those levels produce at fixed $\eta_\oplus$ (red).

Stark et al. (2024) Fig. 10



This model

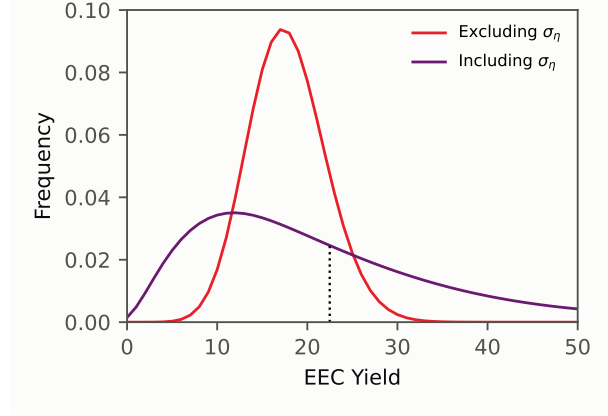
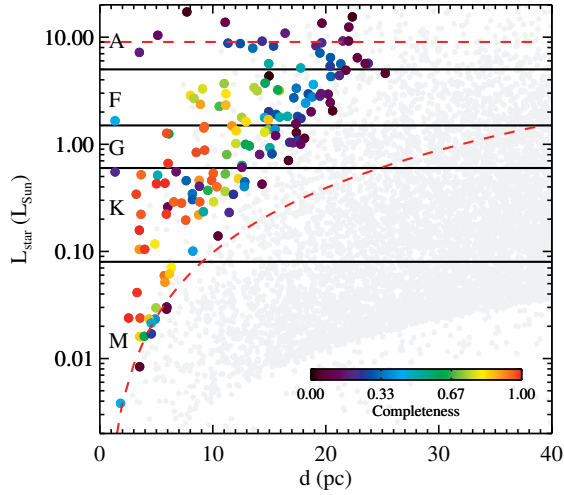


Figure 10: Realized yield including (purple) and excluding (red) the uncertainty in $\eta_{\oplus}$; the dotted line marks the mean of the purple curve. Total-variation distance between model and published curves: 0.027 (purple) and 0.052 (red). Means (purple / red): published 21.0 / 17.3, model 22.5 / 17.8.

Stark et al. (2024) Fig. 11



This model

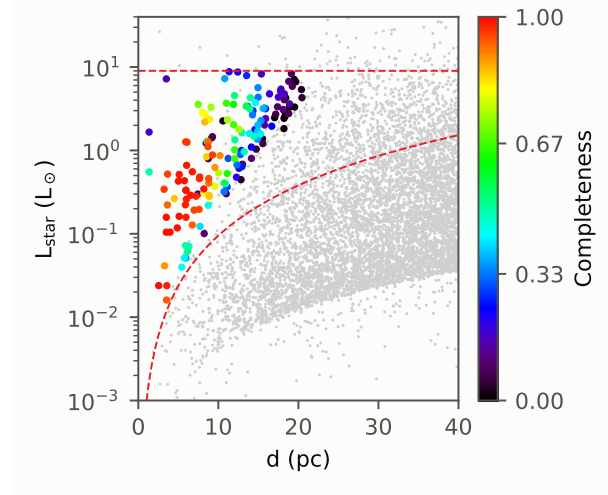
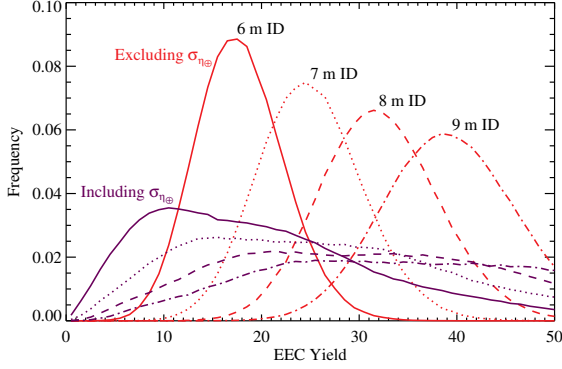


Figure 11: Targets selected in one representative exozodi draw, colored by completeness and plotted over the full input list (gray). Both panels use Stark's own color scale, read from the color bar embedded in the published figure. Published: 154 targets, summed completeness 77.4, median 0.50. Model (draw 14, the draw with the median yield): 157 targets, summed completeness 77.0, and median 0.09 over the 141 targets that match. The red dashed lines are the noise floor for a $1.4 R_{\oplus}$ planet at the EEID, and the outer edge of the habitable zone at $1.5 \lambda/D$ at $1 \mu\text{m}$. The spectral-type lines are not redrawn.

Stark et al. (2024) Fig. 12



This model

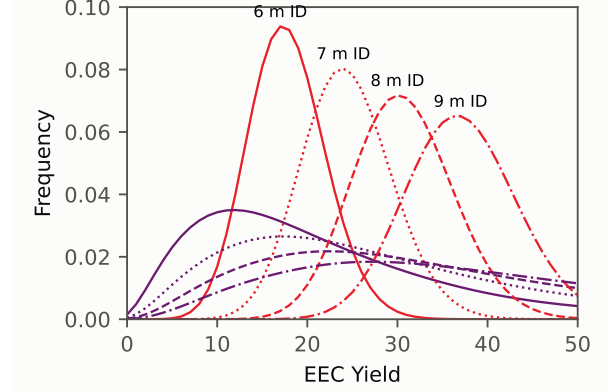
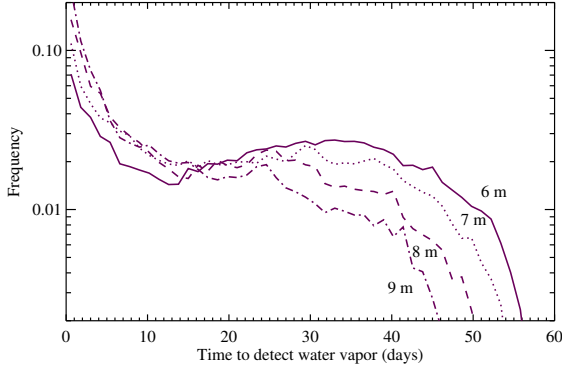


Figure 12: Realized yield at inscribed diameters of 6, 7, 8 and 9 m (solid, dotted, dashed and dot-dashed), including (purple) and excluding (red) the uncertainty in $\eta_{\oplus}$, with no calibration at any aperture. Excluding that uncertainty, Stark’s 9 m yield is 2.2 times his 6 m yield; the model’s is 2.09 times.

Stark et al. (2024) Fig. 14



This model

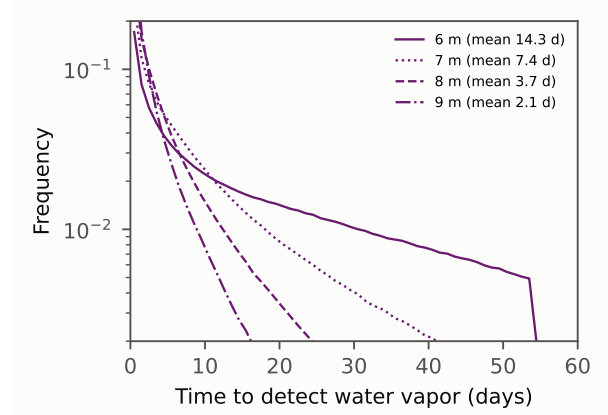
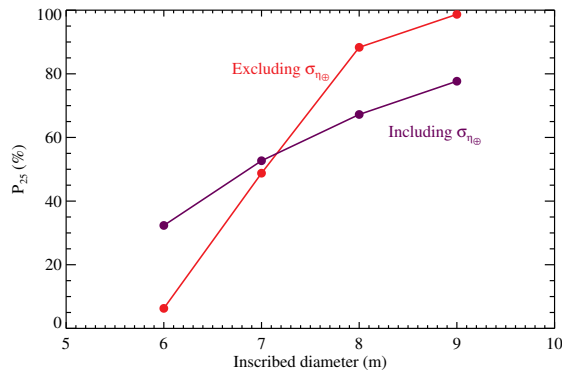


Figure 13: Characterization times of the individual EECs among the first 18, at 6–9 m. The model uses 20 exozodi draws, recomputed planet by planet, in 1-day bins. Published means: 22 days at 6 m and 3.5 days at 9 m. Model means at 6/7/8/9 m: 14.3 / 7.4 / 3.7 / 2.1 days. Stark does not state how his histograms are normalized, and with 1-day bins his 9 m plateau alone would exceed his stated mean, so compare the shapes rather than the heights. His distributions keep a plateau of 15–50-day spectra at every aperture; the model’s do not.

Stark et al. (2024) Fig. 15



This model

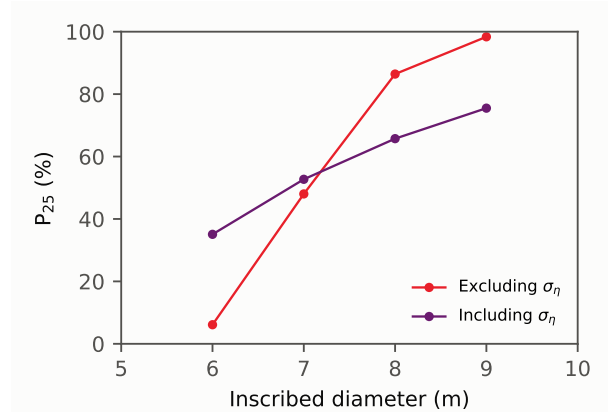
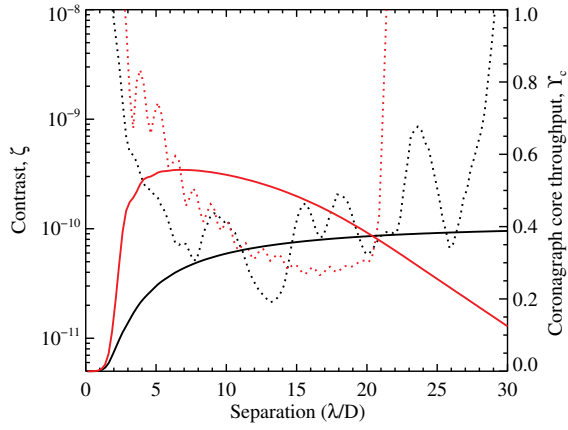


Figure 14: P_{25} , the probability of at least 25 candidates, against inscribed diameter (6/7/8/9 m). Including the uncertainty in $\eta_{\oplus}$: published 32/53/67/78%, model 35 / 53 / 66 / 76%. Excluding it: published 6/49/89/99.5%, model 6 / 48 / 86 / 98%.

Hypothesis	Outcome	Evidence
Missing revisits	Real, a cause	Single-visit only; modeling up to six visits raised the uncalibrated yield $1.7\times$.
Characterization-time statistic	Real, a cause	A median over all detectable planets, where the budget needs a mean over those counted; $D^{1.11} \rightarrow D^{1.88}$.
Over-cap characterization charged	Real, a cause	15 000 days charged at 20 pc for an observation never attempted.
Reconstructed target catalog	Real, a cause	HPIC was reachable; the Gaia stand-in lacked 26–34% of distant FGK dwarfs and ran hot.
Coronagraph aperture convention	Real, the cause	Curves are in circumscribed λ/D ; the IWA sat 19% too far out. Yield $14.7 \rightarrow 18.7$.
Collecting area as inscribed circle	Real, partial	Charged the Lyot discard twice.
Characterization at detection phase	Real, a cause	Stark (2019) Sec. 5: the orbit is known, so “the phase of the planet could be optimized.
Characterization budget under albedo draw	Real, a cause	Stark (2024) Sec. 3.2 fixes it at $A_G = 0.2$; albedo penalty $0.050 \rightarrow 0.087$.
Constant T_{sky}	Real, the cause of κ	Stark (2019) Eqs. 5–6 make it a map; with it the uncalibrated yield is 20.92 and κ falls $1.62 \rightarrow 0.987$. Characterization times barely move.
Fitted throughput factor κ	Real, removed	22.5 is an output of AYO; once refitted, κ turned every tested lever into a near-uniform 1.0 ± 0.05 change in exposure times.
One exozodi draw	Real, fixed	Now 200 draws with the survey re-optimized per draw; exozodi penalty 12.2% (Stark 11.1%).
Split κ (detection vs spectra)	Rejected	A scalar on spectra cannot change the 6/9 m characterization-time ratio.
Exozodi drawn during calibration	Real, a cause	κ fitted to a drawn-exozodi yield cancels the exozodi penalty.
Lognormal exozodi stand-in	Real, a cause	0.2% of stars above 100 zodis vs 21% in HOSTS; penalty $0.7\% \rightarrow 12.2\%$.
Pinned LBTI stars omitted	Real, partial	Worth at most 0.47 EEC here.
$\eta_{\oplus}$ interval read as 86%	Real, the cause of the shape	Fig. 10 requires ln-width ~ 0.8 ; distance $0.142 \rightarrow 0.024$.
Published mode ill-determined	Rejected	Stark’s 498 runs carry 1000 planet draws each; an earlier bootstrap resampled only 498 yields.
Fig. 4 mean compared with the wrong yield	Real, not the cause	Fig. 4 is sampling only at $A_G = 0.2$; the check is independent now that nothing is fitted.
T_{sky} missing; exozodi law	Real, not the cause	Backgrounds $1.48\times$ too bright; constant-surface-brightness law. Together 1%.
Contrast floor inside the knee	Real, not the cause	Floor applied universally.
Noise-floor screen	Real, not the cause	Admitted nothing above $4.6 L_{\odot}$.
Screening cap, uniform bias, band-pass choice	Rejected	1.3%, destroys 7–9 m, 0.01.
Two detections for orbits	Rejected	Shifts the curve, does not narrow it.
Equal-slope allocation	Rejected	0.00% from an exact optimum.
Blackbody stellar flux	Rejected	+0.019 mag against measured V .
Re-derived exposures at the drawn albedo	Replaced	Stark’s per-visit test, both bounds, is adopted; it contributes 0.9% of the Fig. 10 disagreement and gives a penalty of 3.6% (published 12%).
Noise floor, CIC definition, IFS pixels, ordering, dust color	Rejected	Sec. 5.

Table 7: The ledger. The traps that recur — compensating errors, comparing against the wrong published curve, biased diagnostic sub-samples, and read-off values — are documented in `doc/handoffToNextAgent.md`.

Stark et al. (2024) Fig. 25



This model

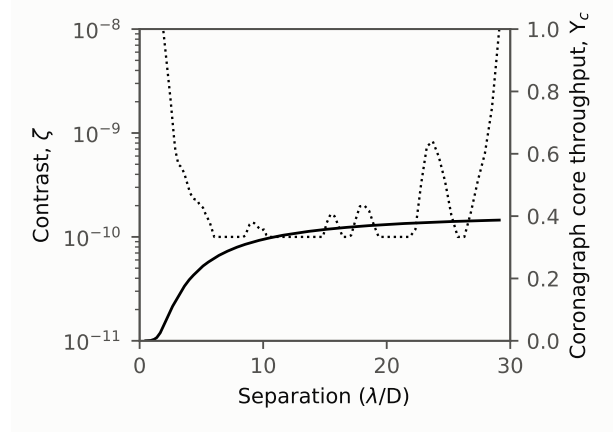


Figure 15: DMVC6 raw contrast (dotted) and core throughput (solid) against separation in circumscribed λ/D . The model curves are the digitized published ones, evaluated exactly as the count rates use them, so this pair checks the digitization, not the model. The model's contrast is drawn with the 10^{-10} floor applied, as the exposure-time calculation uses it; the published curve is the raw simulation. The published figure also shows the PIAA-FPM2.5 coronagraph (red), which is not used here.